



FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Füst
Prof. Dr.-Ing. Florian Risch

**Institute for Factory Automation
and Production Systems**



Friedrich-Alexander-Universität
Technische Fakultät

Production systems for electric drive technology

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

Based on current challenges, approaches to reducing costs, increasing scalability, using Industry 4.0 technologies and recycling are outlined.

Producers of electric drive systems have to adapt to changing market situations ever more quickly. In order to survive in international competition, they must be able to produce different variants in the shortest possible time with increasing quantities in high quality at a competitive price. Approaches to controlling costs, increasing scalability, reducing product complexity and the targeted use of Industry 4.0 technologies are therefore more important than ever.

After attending the lecture unit "Production systems for electric drive technology", you should

- **outline current challenges and trends in electric motor production,**
- **understand cost structures and master measures for optimizing production costs,**
- **apply solutions for scalable multi-variant production as required, and**
- **assess the potential of Industry 4.0, in particular artificial intelligence, in electric motor production.**



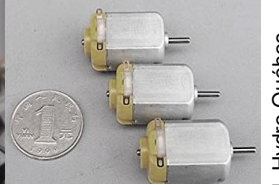
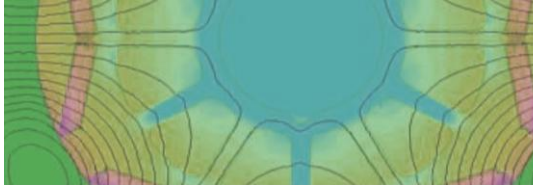
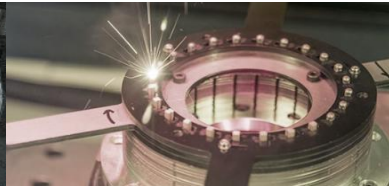
Based on current challenges, approaches to reducing costs, increasing scalability, using Industry 4.0 technologies and recycling are outlined.

- **Current challenges and trends in electric motor production**
- Cost structures and measures to optimize production costs
- Solution approaches for scalable multi-variant production
- Potentials of Industry 4.0 and artificial intelligence



Graphic: VW

The current challenges in the production of electric motors, especially traction motors for electric vehicles, can be divided into five fields of action.

1	Product technologies	Operational requirements and application areas determine the development and selection of suitable technologies	  
2	Quality specification	Performance data and quality requirements influence the selection of production processes	 
3	Production technologies	Industrialization and automation of highly product-dependent production processes are key challenges	  
4	Start-up	In practice, (re)starting currently represents a major challenge for system manufacturers, suppliers and engine builders	  
5	Costs	Target cost specifications according to the areas of application restrict the choice of technology in production	 

Graphics: Continental, Trumpf, BMW, Audi, AllExpress, Ramco Electric Motors, FramePool, Hydro-Québec

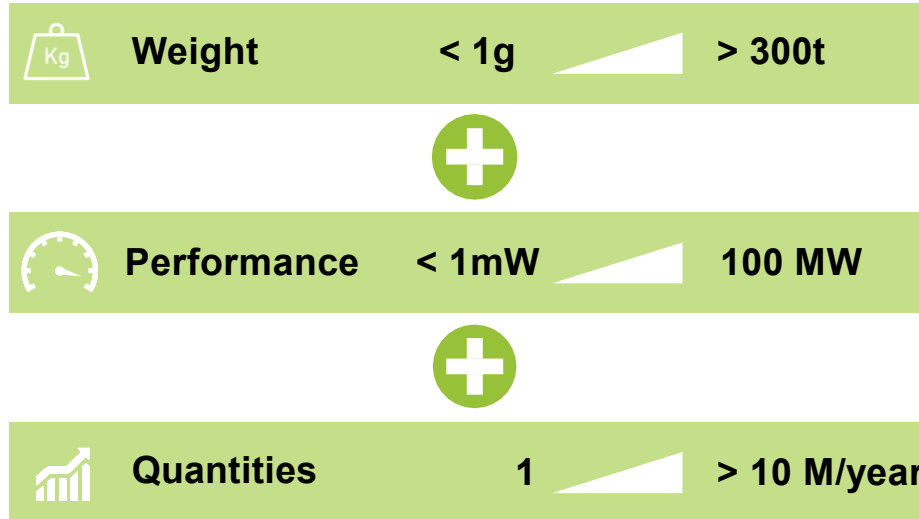
Due to the wide range of applications for electric motors, the electric motor industry is characterized by enormous product diversity and a high degree of individualization.

1	Product technologies
2	Quality specification
3	Production technologies
4	Start-up
5	Costs

Product diversity



Graphic: Sol Expert



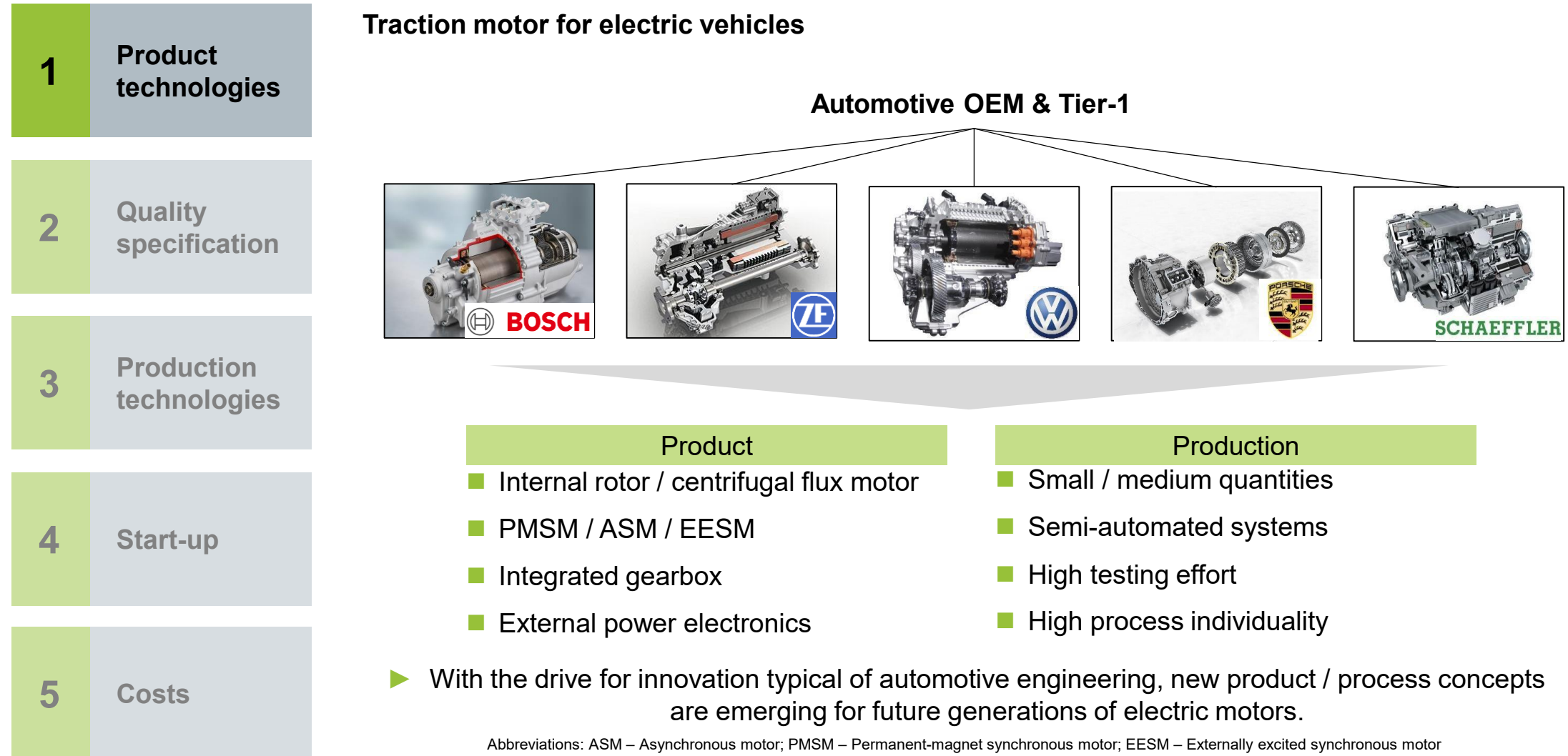
Graphic: Menzel-Motors



Graphic: Continental

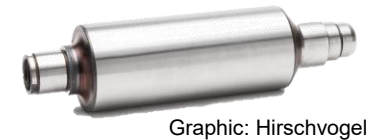
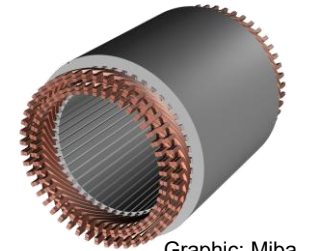
- The use of electric motors presents a very **heterogeneous picture** of different application scenarios.
- Depending on the application, the production of electric motors requires a **high degree of customization**.
- **Few overarching standards** have yet been established in both product and production technologies.

Nevertheless, OEMs and tiers have so far developed similar solutions for the traction motors of tomorrow in terms of products and processes.



The shortcomings of the first generation of traction motors produced in-house in the automotive industry have resulted in new product innovations.

1	Product technologies	Traction motor for electric vehicles		
		Stator	Deficits of the first generation(s)	Product innovations
			- High manual switching effort with retracted winding - Long process times - Low fill factor - Low process capability	- Shaped coil concepts: - Hairpin - I-Pin - Shaft winding - Optimization of the degree of automation and product efficiency
		Rotor	- High active material costs due to magnetic material - Low efficiency due to low speed - Heat development with ASM	- Reduction of the NdFeB magnet material or substitution by ferrite magnets - Integration of rotor cooling concepts through hollow shafts
			- Complex contacting - Long process times - Low process capability	- Simplified and faster assembly using switching or contact rings - Realization of more complex winding schemes - Complete system integration (motor, gearbox, electronics, cooling)
2	Quality specification			
3	Production technologies			
4	Start-up			
5	Costs			



Cost savings on the user side can be achieved through the integration of peripheral components.

1

Product technologies

2

Quality specification

3

Production technologies

4

Start-up

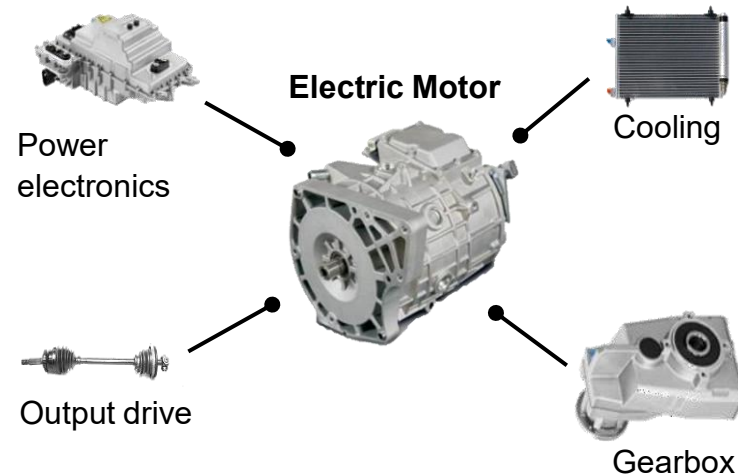
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Costs

Highly integrated systems as a future product scenario

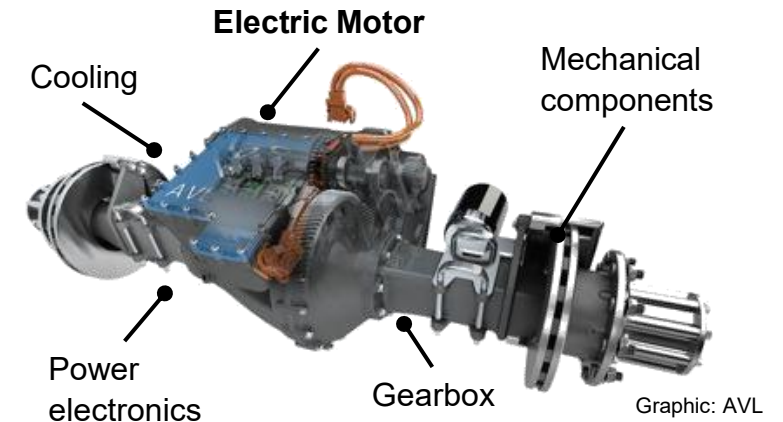
Functional integration is currently being pursued to reduce costs and weight, particularly in weight-critical applications such as the automotive industry.

System of individual components



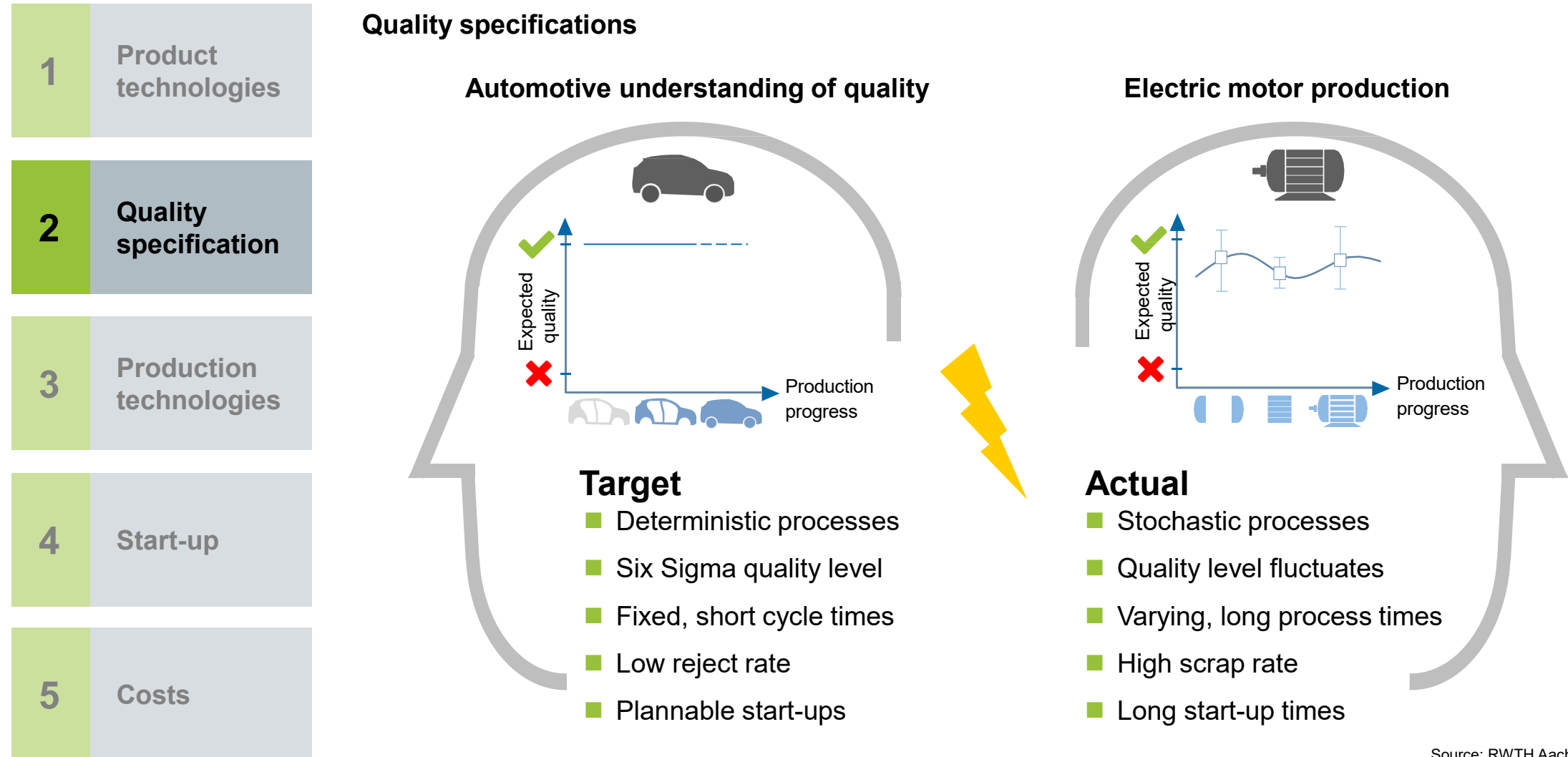
- High number of variants
- Risk of incompatibility
- Large space requirement
- Distributed value chain

Integrated axle



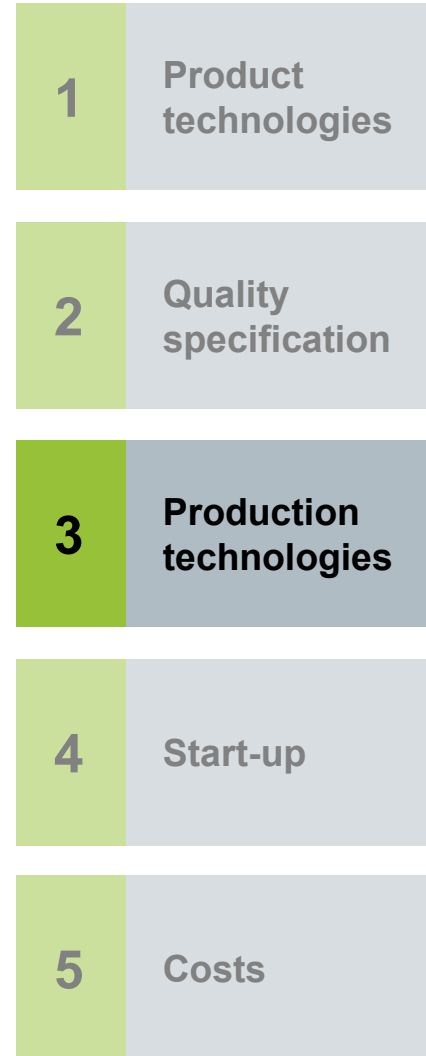
- Complete system integration, resulting in lower space requirements and reduced weight
- Simplified final assembly, as the number of individual components is reduced
- Shift in the value chain

The previous, stochastic processes of established electric motor production for industrial motors are in conflict with the automotive understanding of quality.

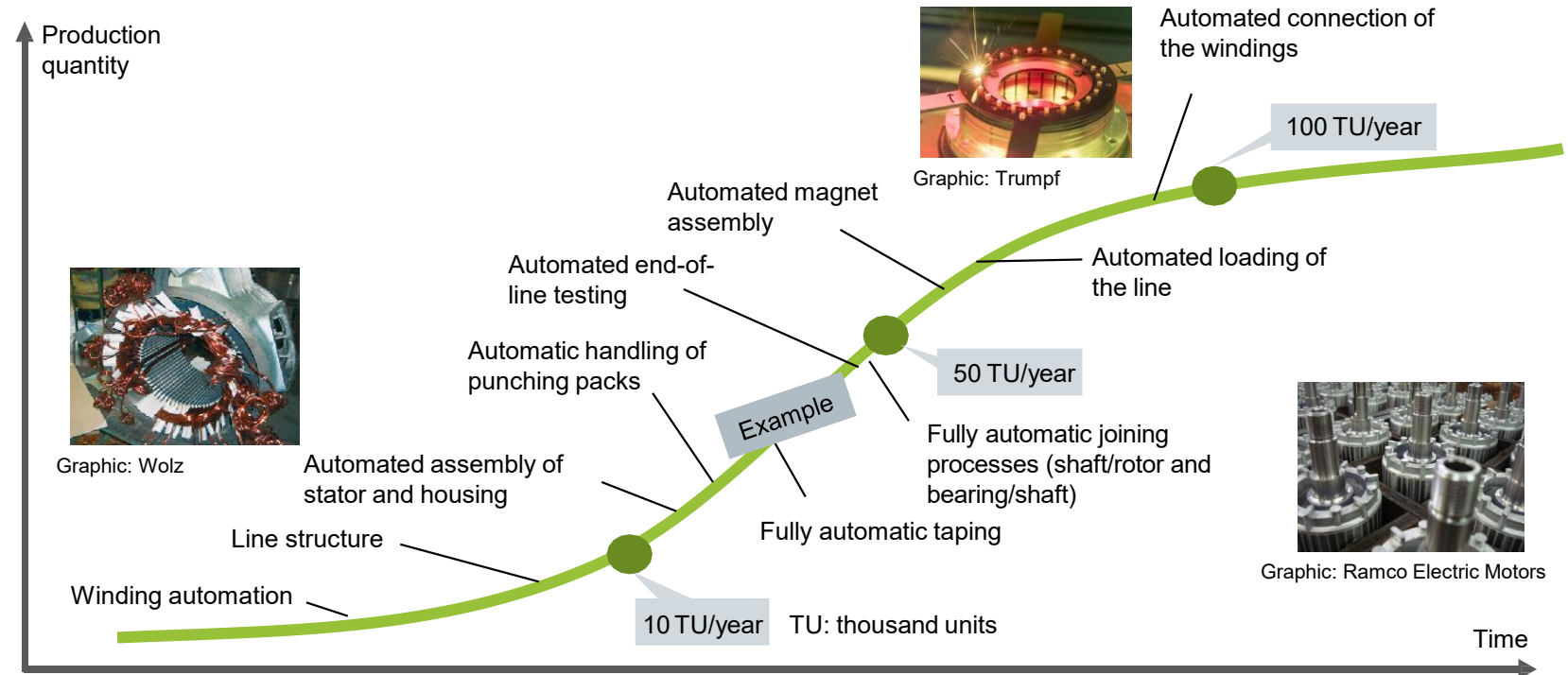


Source: RWTH Aachen

The appropriate degree of automation must be selected depending on the planned output.



Exemplary degrees of automation in electric motor production



- Process reliability is a key challenge in the industrialization and automation of efficient production; in wire winding in particular, bending slack makes it difficult to develop reliable automation solutions
- Competitive advantages can be gained through automation technology suitable for large-scale production, among other things

Source: RWTH Aachen

The start of mass production is currently presenting many electric motor manufacturers with major challenges.

1

Product technologies

Challenges in the run-up

- Unbalanced requirements specification towards suppliers and equipment manufacturers
- Unforeseen process interdependencies
- Granularity of test frequencies inside and outside the line
- Lack of reference systems
- High degree of automation and system interlinking in high-volume production

2

Quality specification

3

Production technologies

4

Start-up

5

Costs

Alongside the battery, the drive system of an electric vehicle is a major cost factor in automobile production.

- 1 Product technologies
- 2 Quality specification
- 3 Production technologies
- 4 Start-up
- 5 Costs

Cost structures

Significantly fewer parts

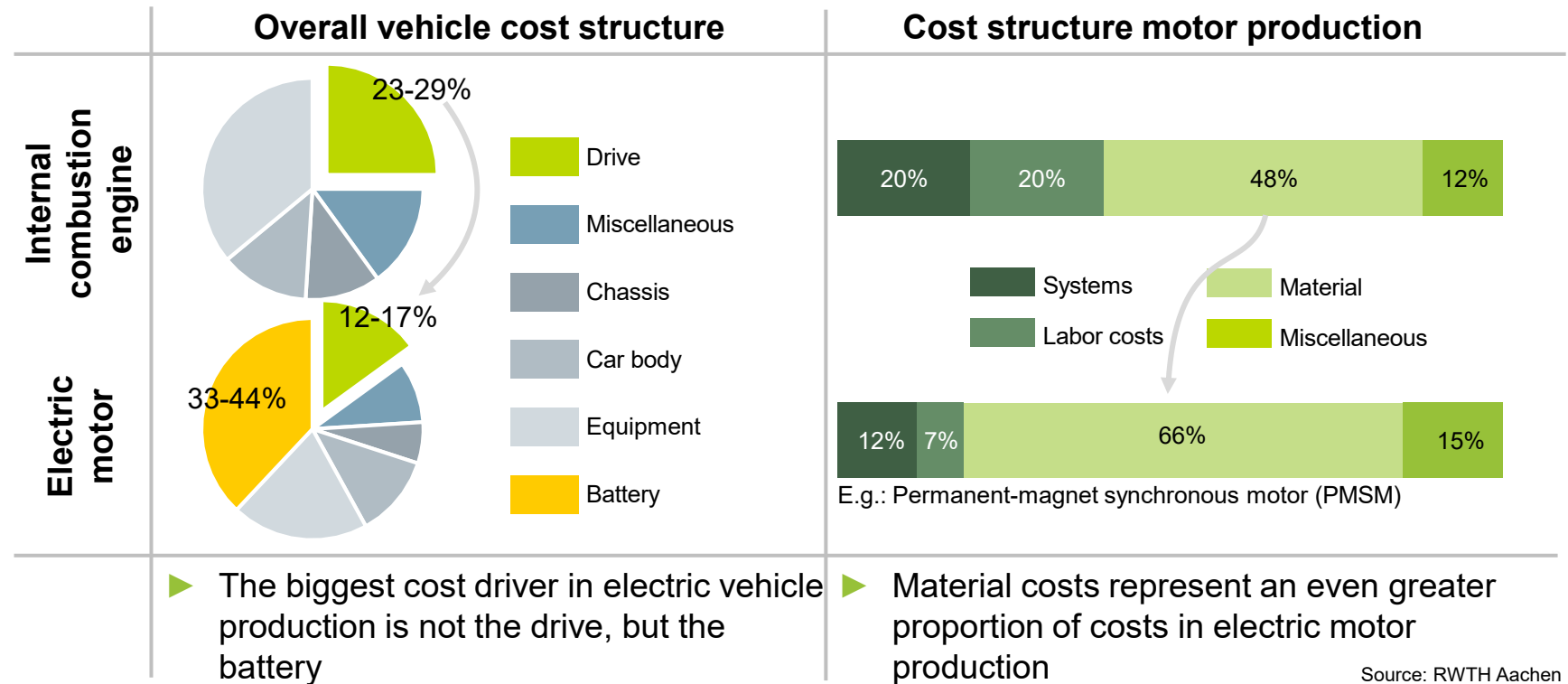


Combustion engine
1,400 parts*

Electric motor
210 parts*

Significant changes lead to a reorganization of value creation

*Estimated values

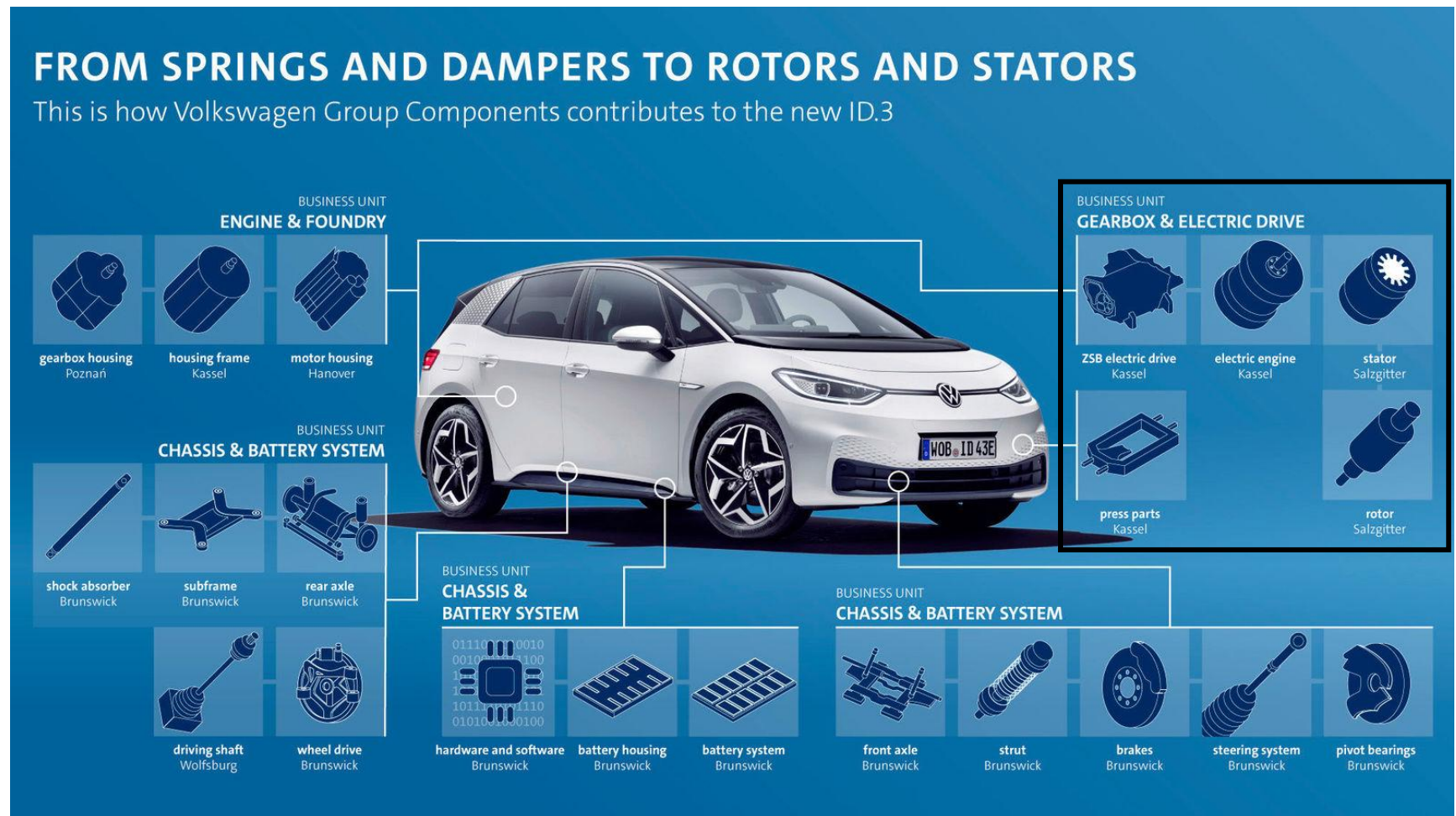


Source: RWTH Aachen

The VW Components plants produce in a mix of location specialization with simultaneous integrated production in order to leverage synergies and use specific know-how.

- 1 Product technologies
- 2 Quality specification
- 3 Production technologies
- 4 Start-up
- 5 Costs

Example Volkswagen



Graphic: VW

Based on current challenges, approaches to reducing costs, increasing scalability, using Industry 4.0 technologies and recycling are outlined.

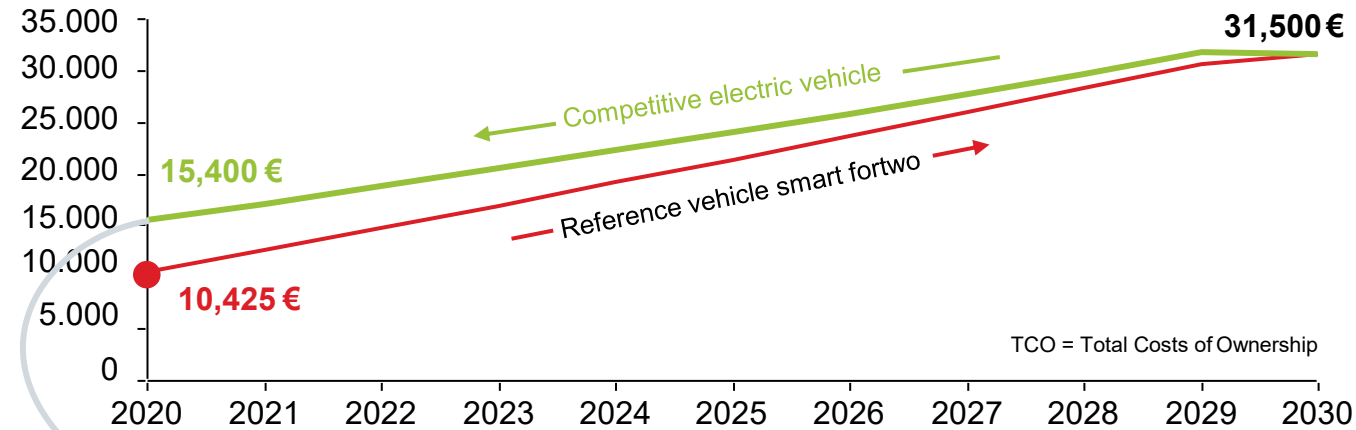
- Current challenges and trends in electric motor production
- **Cost structures and measures to optimize production costs**
- Solution approaches for scalable multi-variant production
- Potentials of Industry 4.0 and artificial intelligence



Graphic: VW

The target production costs for a competitive electric vehicle are based on a TCO analysis of an exemplary conventional subcompact car.

Exemplary competitive sales price for a small electric car based on the TCO



Smart fortwo (2020)



Price: 10.425 €
TCO: 31.500 €

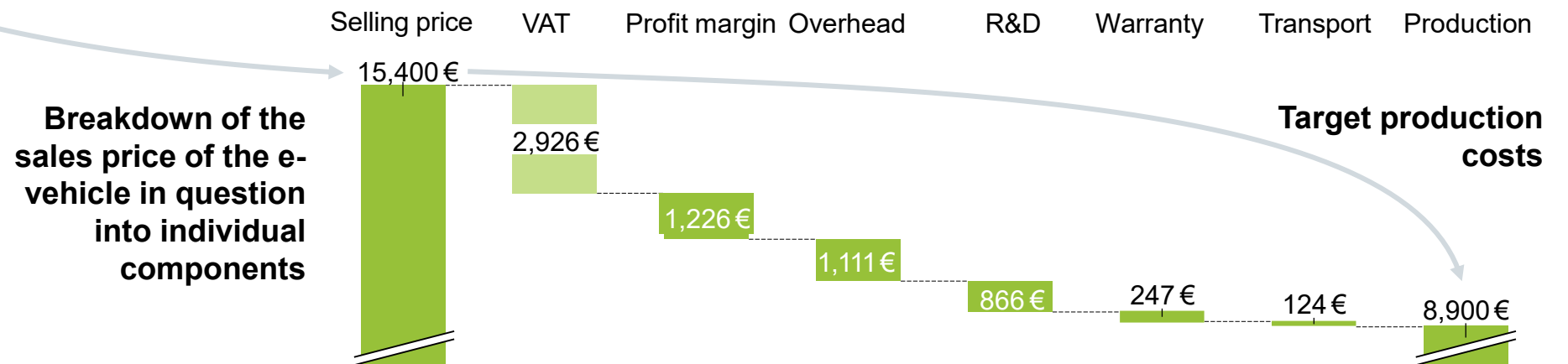
Small electric car (target)



Price : 15,400 €
TCO: 31,500 €

Assumptions:

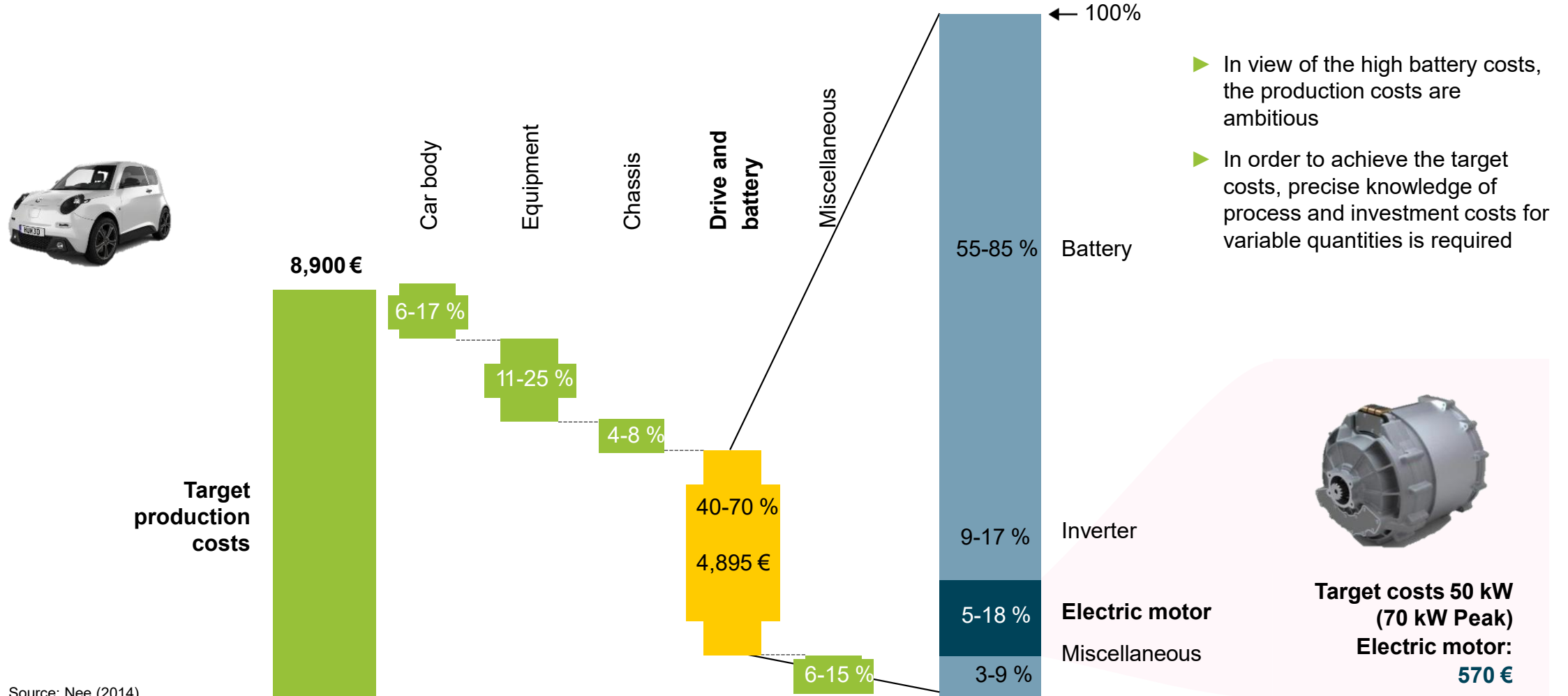
	Smart	BEV
Annual mileage	15,000 km	
Energy costs	1.40 €/l	28 ct/kWh
Dynamic energy cost	1 %	4 %
Consumption/100 km	4,5l	14 kWh
Taxes/year	38 €	0 €
Insurance/year	469 €	
Workshop costs/year	396 €	300 €
Operating costs/year	160 €	120 €
Cost of capital (Opport.)	1 %	
Residual value	1,472 €	2,174 €



Source: Nee (2014)

The target costs of the traction motor are derived from the calculated target costs of producing the small electric vehicle under consideration.

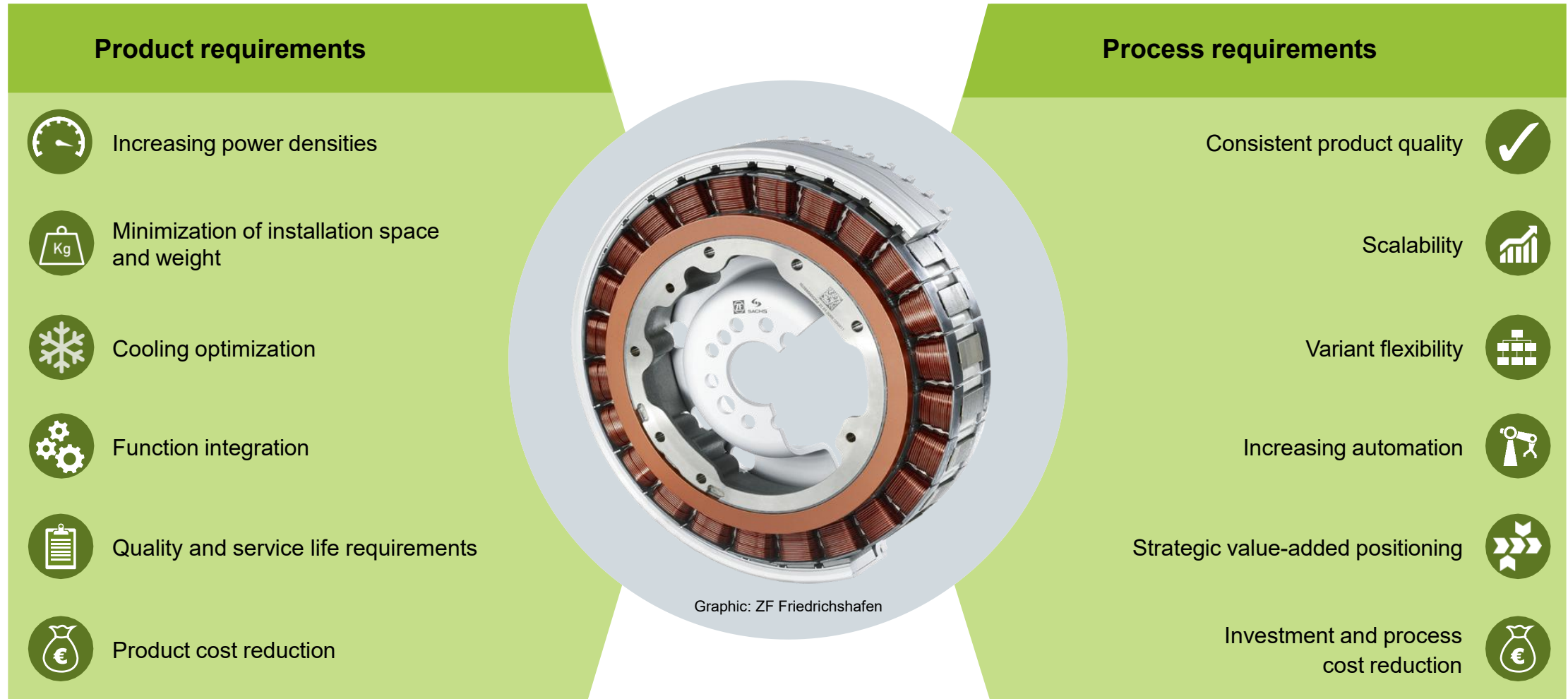
Example calculation of target costs for the production of an electric motor



Source: Nee (2014)

Cost targets cannot be viewed in isolation from the sometimes conflicting product and process requirements.

Slide from Lec. 1



There are many possible measures for optimizing costs.



Specification optimization



**Variant management /
product construction
kits**



**Supplier as a source of
innovation**



**Investment cost
analysis**



**Scalable production
system**

...

...



Graphic: Elektror airsystems

The average operating time of a traction motor for electric vehicles is significantly shorter than that of a pump or fan motor, which are designed for continuous operation.



Specification optimization



Variant management/product construction kits



Supplier as a source of innovation



Investment cost analysis



Scalable production system

...

...

Application-oriented requirements specification using the example of a traction motor

- Driving cycle and service life-oriented design
- Average operating time of a traction motor for passenger cars approx. 8,000 hours
- Any oversizing in the design leads to an increase in costs
- Industrial motors with a comparable performance class are designed for a significantly longer service life
- Starting points:
 - Dimensioning of the drifting components (ball bearing, shaft, gearbox) to meet requirements
 - By optimizing the insulation class of the enameled wire, stator impregnation and joints, cost potentials can be exploited

Traction in an electric vehicle



Graphic: Siemens

Water pump



Graphic: Wilo

A large number of variants, for example among suppliers of industrial motors, can be mastered by introducing modular product kits.



Specification optimization



Variant management/product construction kits



Supplier as a source of innovation



Investment cost analysis



Scalable production system

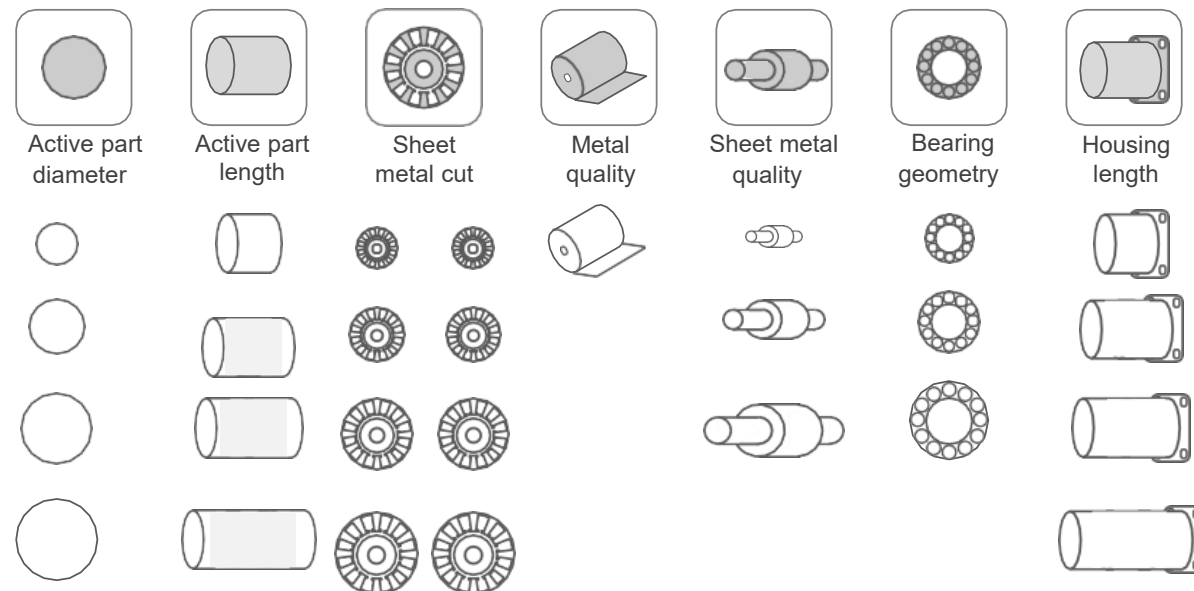


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Procedure for creating modular product kits

- Development of product and process morphology
- Definition of variance-sensitive process steps and critical product features
- Identification and hierarchization of constituent features
- Definition of the product architecture and conceptualization of the process toolkit

Exemplary standards and staging in a modular e-motor system:



Quelle: RWTH Aachen

The great innovation potential of many suppliers is reflected in their high R&D ratio.



Specification optimization



Variant management/product construction kits



Supplier as a source of innovation



Investment cost analysis



Scalable production system

...

...

R&D ratios in the automotive industry

Advantages:

- Utilization of the supplier's pool of ideas
- Avoidance of additional machine investments through the use of existing systems
- Ideas for product adaptation for more favorable production technologies (avoidance of over-engineering)

Prerequisites:

- Integration of purchasing in innovation planning
- Defined supplier development programs

Excerpt from
„The 2018 Global
Innovation 1000 study”
von strategy&

Legend: R&D intensity = R&D expenditures as a percentage of sales

				R&D Intensity
Company Name	Country	Industry group	2018	
Ferrari N.V.	Italy	Automobiles and Components	19.2%	
Brilliance China Automotive...	Hong Kong	Automobiles and Components	15.8%	
Tesla, Inc.	United States	Automobiles and Components	11.7%	
HELLA GmbH & Co. KGaA	Germany	Automobiles and Components	10.0%	
DENSO Corporation	Japan	Automobiles and Components	9.9%	
Delphi Technologies PLC	United Kingdom	Automobiles and Components	8.7%	
Haima Automobile Group C...	China	Automobiles and Components	8.5%	
Visteon Corporation	United States	Automobiles and Components	8.0%	
Autoliv, Inc.	Sweden	Automobiles and Components	7.1%	
Continental Aktiengesellschaft	Germany	Automobiles and Components	7.1%	
Aptiv PLC	Ireland	Automobiles and Components	6.8%	
Jiangling Motors Corporatio...	China	Automobiles and Components	6.6%	
Keihin Corporation	Japan	Automobiles and Components	6.5%	
Valeo SA	France	Automobiles and Components	6.1%	
Schaeffler AG	Germany	Automobiles and Components	6.0%	
Yamaha Motor Co., Ltd.	Japan	Automobiles and Components	5.9%	
Volkswagen Aktiengesellsc...	Germany	Automobiles and Components	5.7%	
Tokai Rika Co., Ltd.	Japan	Automobiles and Components	5.5%	
Honda Motor Co., Ltd.	Japan	Automobiles and Components	5.4%	
Nexteer Automotive Group ...	United States	Automobiles and Components	5.3%	
Aisin Seiki Co., Ltd.	Japan	Automobiles and Components	5.1%	
Ford Motor Company	United States	Automobiles and Components	5.1%	
General Motors Company	United States	Automobiles and Components	5.0%	
Bayerische Motoren Werke ...	Germany	Automobiles and Components	5.0%	
Isuzu Motors Limited	Japan	Automobiles and Components	5.0%	

An investment cost analysis serves to reduce investment costs, for example through low-investment reference process chains or a suitable product design.



Specification optimization



Variant management/product construction kits



Supplier as a source of innovation



Investment cost analysis



Scalable production system

...

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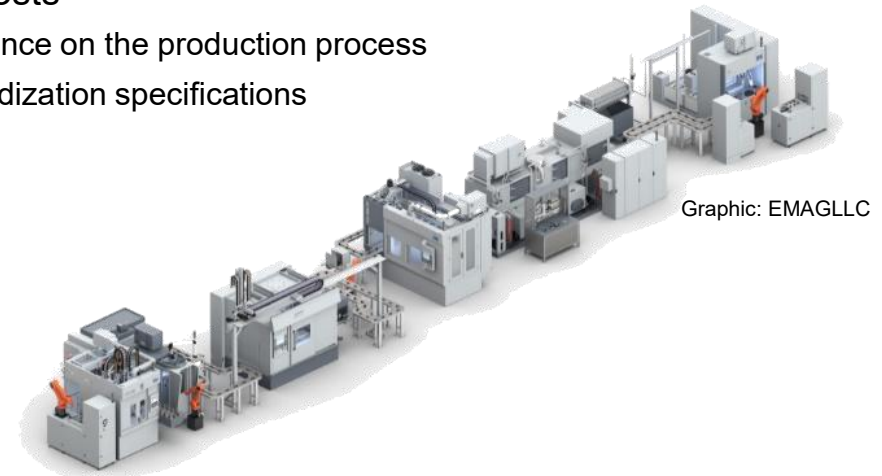
Low-investment process chains

Goal:

Reduction of investment costs for electric motor production

Procedure:

- Low-investment reference process chains for product groups
 - Evaluation and selection of process alternatives in the production process
 - Overview of economical production technologies
 - Increased process flexibility and scalability to enable cross-product life cycle plant utilization
- Product design to reduce investment costs
 - Analysis of product features and their influence on the production process
 - Clustering of products according to standardization specifications
- Deriving a procedure for planning low-investment production lines
- Organizational anchoring of integrated product and process development



Source: RWTH Aachen

Based on current challenges, approaches to reducing costs, increasing scalability, using Industry 4.0 technologies and recycling are outlined.

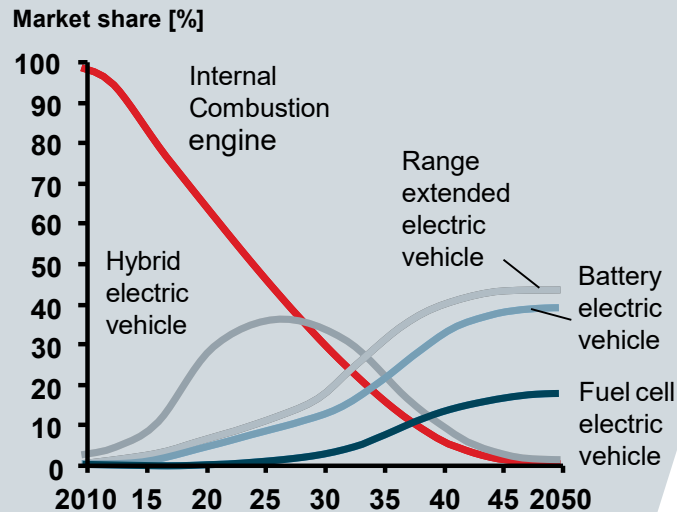
- Current challenges and trends in electric motor production
- Cost structures and measures to optimize production costs
- **Solution approaches for scalable multi-variant production**
- Potentials of Industry 4.0 and artificial intelligence



Graphic: VW

The market for electric traction motors is characterized by enormous diversity in the product portfolio and uncertain sales forecasts.

Uncertainties on the sales market



Source: McKinsey

Challenges

1 Scalability

Due to the increasing emission requirements and the competition between different drive technologies, it is difficult to make unit forecasts.

2 Multivariant production

Despite similar applications, customized motors are sometimes produced for specific customer requirements.

Diversity in the product portfolio

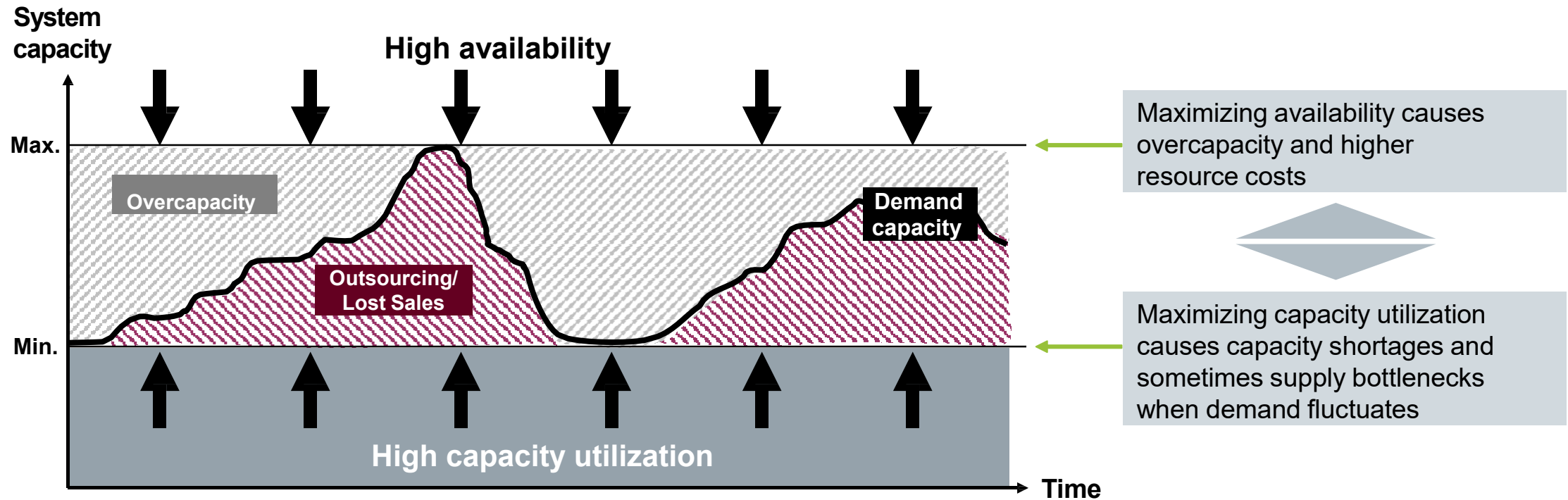
	Renault Twizy	13 kW Synchronous machine
	Smart FORTWO e	35 kW Synchronous machine
	Mitsubishi i-MIEV	49 kW Synchronous machine
	BMW i3	125 kW Synchronous machine
	BMW i8	96 kW Synchronous machine
	Tesla Model S	235-515 kW Asynchronous machine

In order to realize **economies of scope and economies of scale, scalability and multi-variant production** must be taken into account as early as the design stage of a production system

Sources: RWTH Aachen based on Renault, Mia, Smart, Citroen, Mitsubishi, Tesla; Westkämper & Zahn (2009)

One of the most important issues in capacity planning is how to deal with fluctuating demand.

Classic capacity planning as a one-off strategic decision

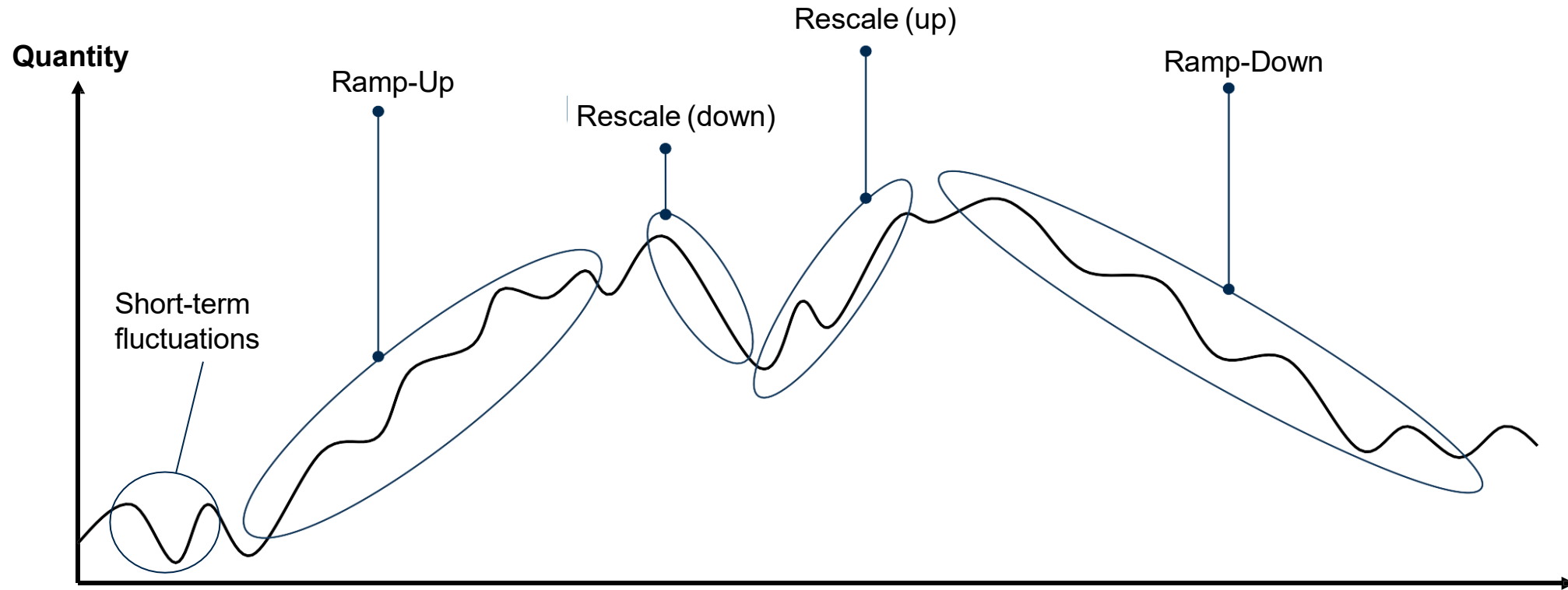


In **traditional approaches**, capacity planning is seen as a **one-off strategic decision**.
The associated over- and undercapacities influence profitability.

Source: RWTH Aachen

Flexibility measures must enable reactions to various changes in sales.

Sales over a product life cycle



When designing a production system, different sales change scenarios should already be taken into account and scalability measures planned.

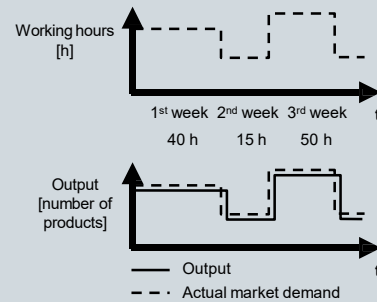
Source: RWTH Aachen

The four levels of scalability must be taken into account for a flexible production system.

Four levels of scalability

1 Personnel scalability

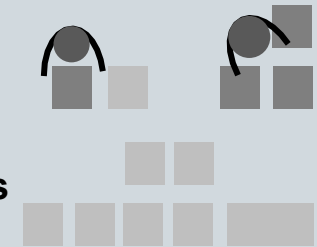
Flexible working hours and shift systems support scalability, even in highly manual production systems. In addition, further training is a long-term measure.



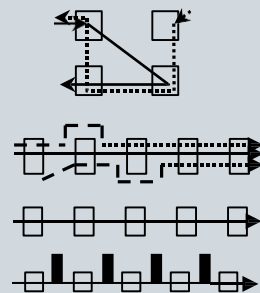
Technical scalability

2

Modular machines with standardized **modules** offer a high degree of scalability at system level. Different cycle times can be achieved through different **degrees of automation**.

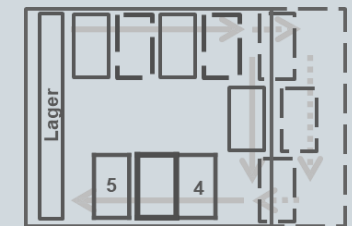


When selecting a suitable **form of production** (e.g. workshop, series or flow production), product and production-related restrictions must be taken into account.



4 Structural scalability

Forward-looking factory planning takes into account technical, personnel and organizational scalability by planning **open spaces** or defining a **growth axis**.



Spatial scalability

3

Source: RWTH Aachen

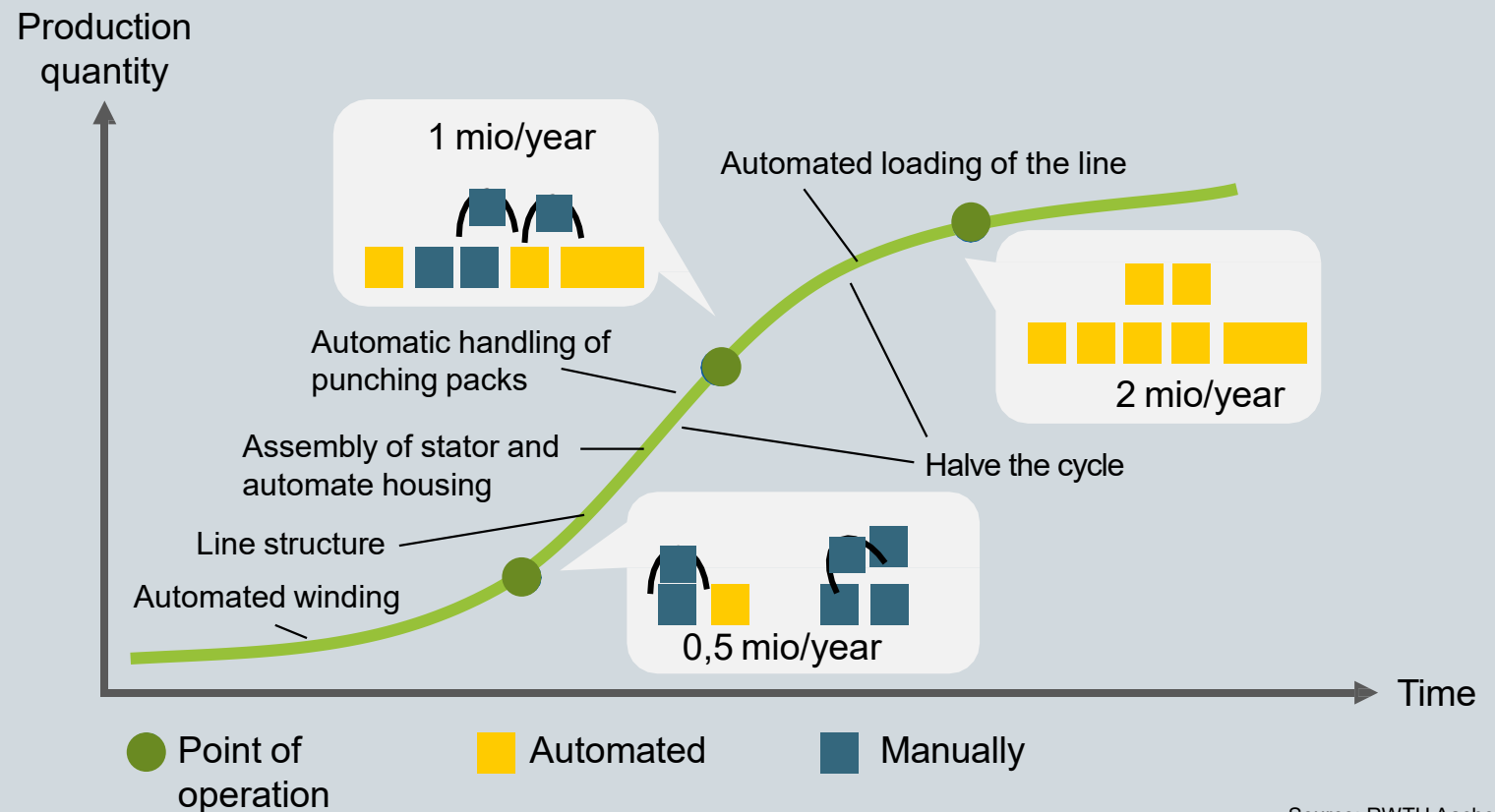
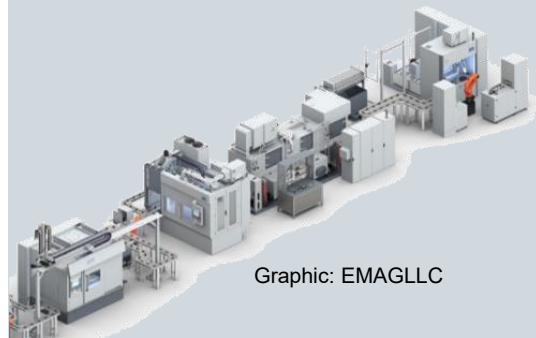
The four levels of scalability must be taken into account for a flexible production system.

Four levels of scalability

Technical scalability

2

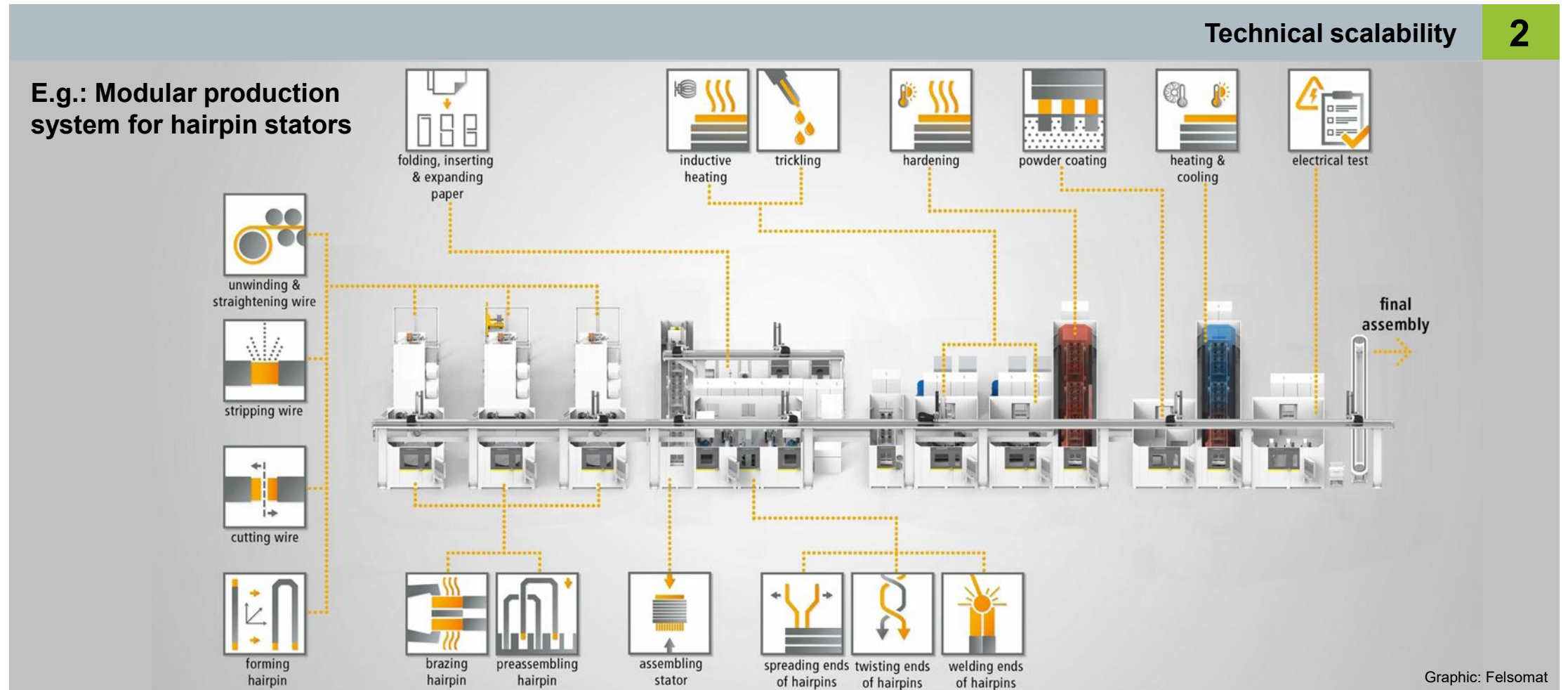
- Technical scalability can be achieved through modularity and different degrees of automation.
- Expandability and the exchange of modules are strong enablers for scalable production systems.



Source: RWTH Aachen

The four levels of scalability must be taken into account for a flexible production system.

Four levels of scalability



The four levels of scalability must be taken into account for a flexible production system.

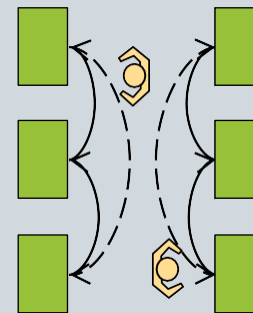
Four levels of scalability

E.g.: Multi-machine operation – Chaku Chaku

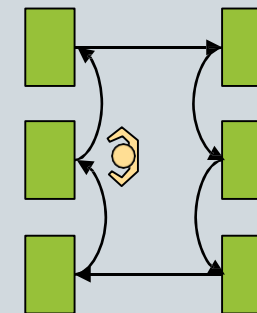


Various forms of production have become established in industrial production. Particularly in ramp-down phases, suitable measures must be taken for efficient structural adjustment.

High utilization



Low utilization



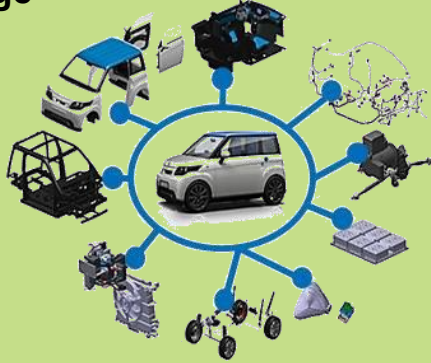
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Structural scalability

The challenges of multi-variant production can be overcome through product and production standardization.

Product standardization and production standardization for scalable multi-variant production

Challenge



- Vehicles, and passenger cars in particular, are extremely complex industrial products
- The drive and therefore the motor is generally not a constitutive feature of the vehicle
- Traction motors must therefore be adapted for different applications, installation spaces, integration levels, etc.

Product standardization

Goal: Standardization of individual parts (standardization) or complete product variants (typing)

Solution:

- Feature standardization
- Industrial modules
- Example: VW's modular e-drive kit (MEB)

Production standardization

Goal: Machines and systems for the production of as many product variants as possible

Solution:

- Variant-flexible machines and systems
- Identical process sequences
- Identical characteristics

The solution is the right **balance between flexibility and standardization**.
Standardized component features are one solution to the balance problem.

Source: RWTH Aachen

While the variance of some product features has little impact on electric motor production lines, others lead to a comprehensive need for flexibility.

Challenges in scalable multi-variant production

Exemplary features with mostly minor effects:

- Active part lengths (rotor / stator)
- Housing length
- Sheet metal cut for joining processes
- Built-in / attached parts (e.g. temperature sensors) for manual processes
- Fits for CNC milling

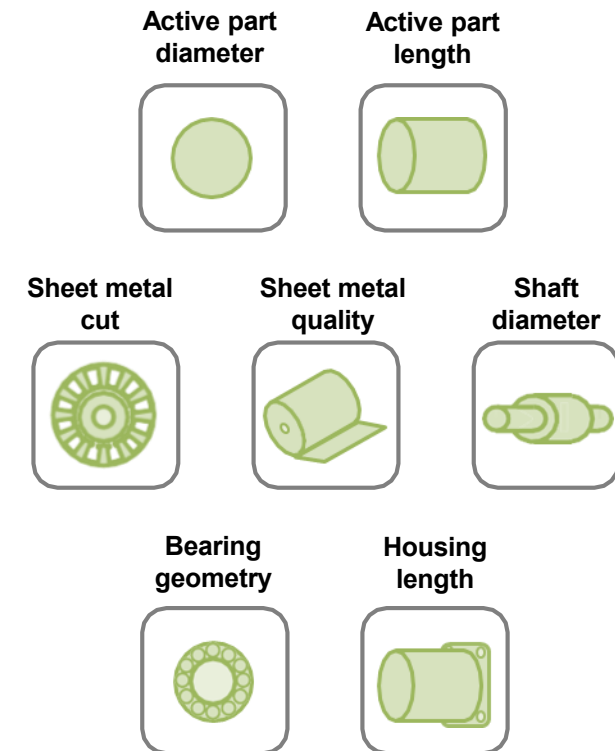
Exemplary features with mostly high impact:

- Active part diameter (rotor / stator)
- Housing material for welded joints
- Sheet metal cut for punching processes
- Different installation / attachment parts for automated Processes
- Wire cross-section shape (round vs. flat)
- Different insulation / resin systems

Mapping the variance by **parameterizing** the production systems

Mapping the variance through **retooling, flexible tools or system changes**

Exemplary product features of an electric motor kit



Source: RWTH Aachen

Based on current challenges, approaches to reducing costs, increasing scalability, using Industry 4.0 technologies and recycling are outlined.

- Current challenges and trends in electric motor production
- Cost structures and measures to optimize production costs
- Solution approaches for scalable multi-variant production
- **Potentials of Industry 4.0 and artificial intelligence**
 - Semantic technologies to support the design of production systems
 - Machine learning for process and production optimization

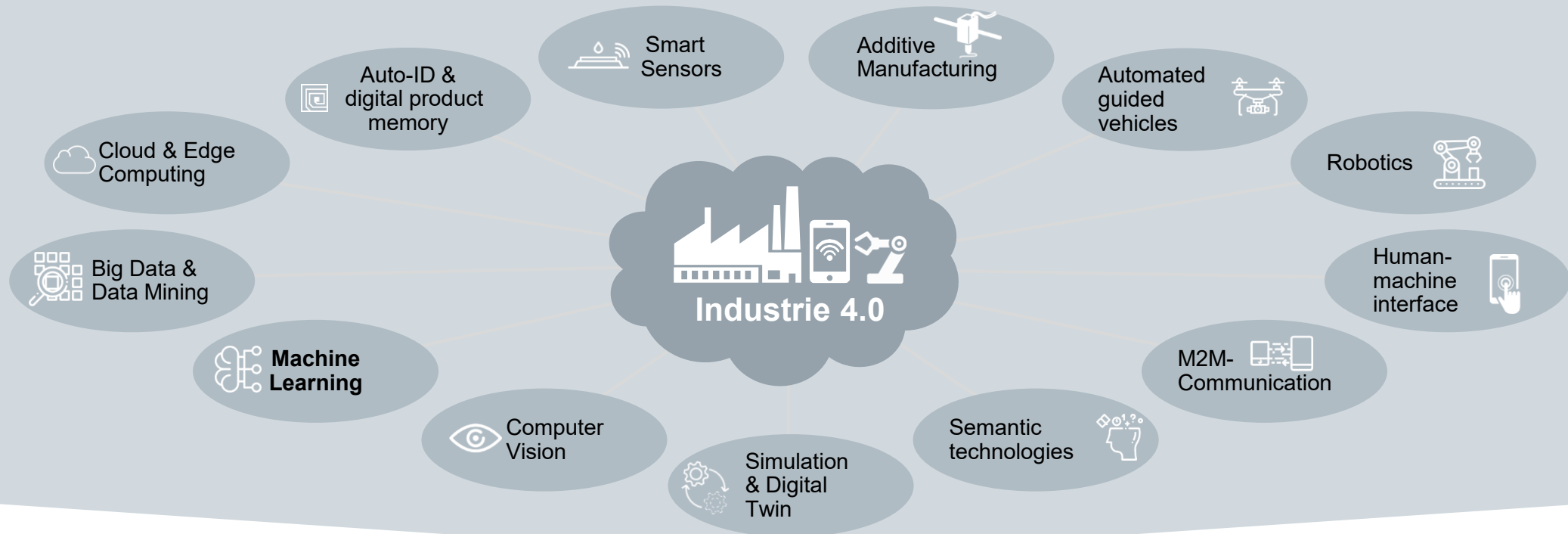


Graphic: VW

Industry 4.0 encompasses a wide range of technologies that also promise great potential in electric motor production.

Definition of industry (I4.0) and underlying technologies

Industry 4.0 refers to the intelligent networking of machines and processes in industry with the help of information and communication technology. (BMW)

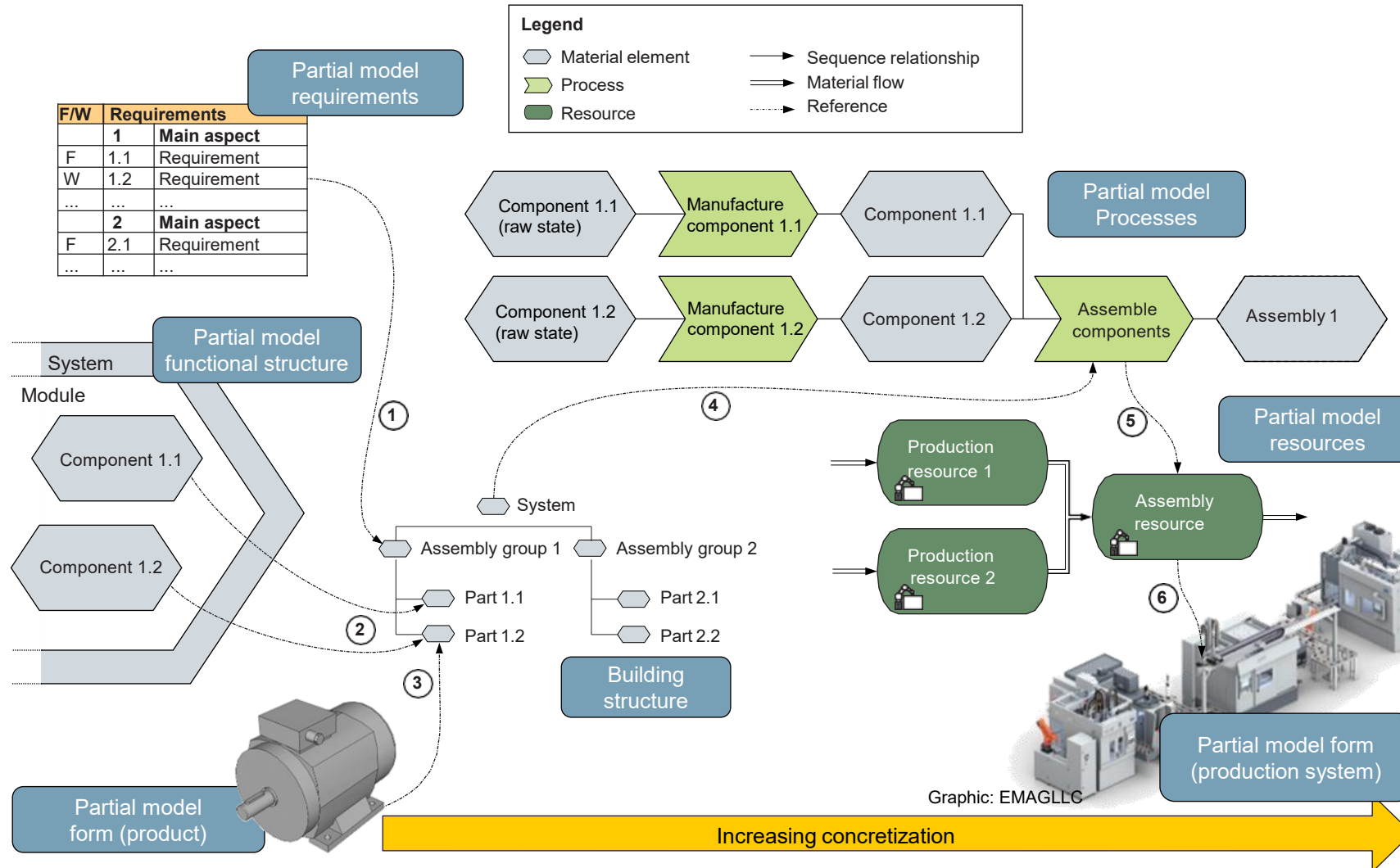


Quelle: Mayr, Weigelt, Lindenfels et al. (2018)



Current research objective: Tapping the potential of Industry 4.0 in electric motor production

The starting point for production system design is the principle solution of the product, on the basis of which the processes, resources and production system design are derived.



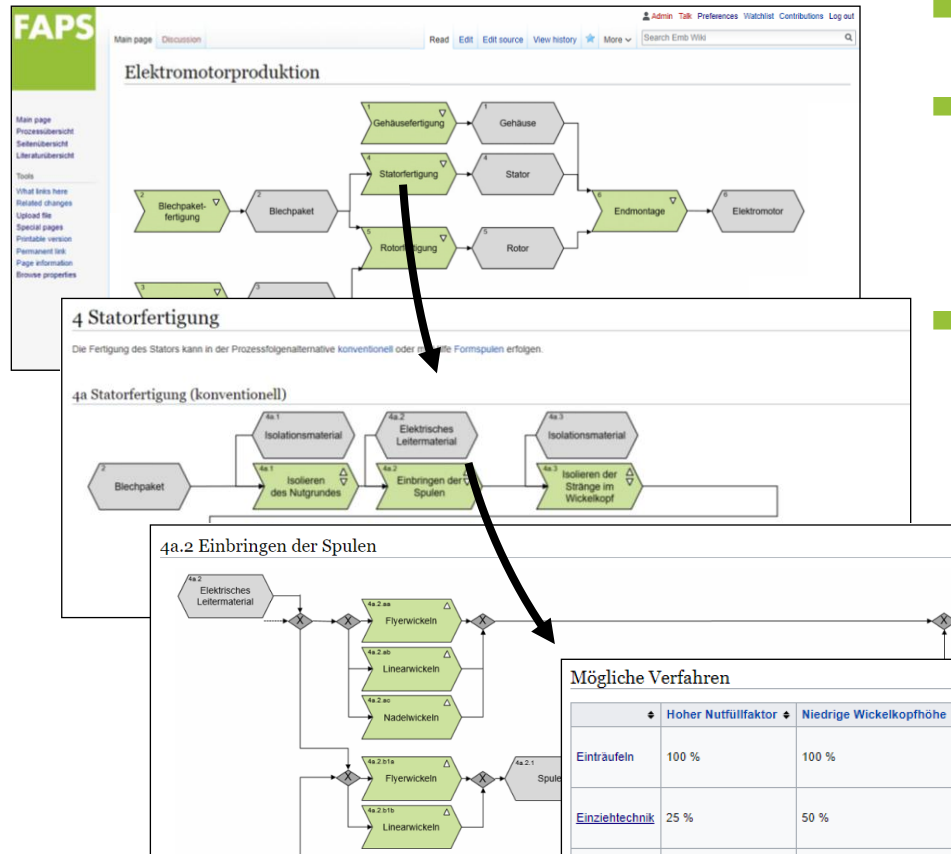
Sequence of production system design:

- Starting from the partial models requirements (1), functional structure (2), and form (3), a manufacturing-appropriate structural design is formed, which serves as the starting point for the subsequent specification of the processes (4).
- Based on the selected processes (5), the necessary resources are chosen, and their form and arrangement are determined (6).

Source: Gausemeier et al. (2012)

Semantic technologies such as semantic wikis can support production planners by providing specialist knowledge and solutions for electric motor production.

Application example | Semantic Wiki to support production system design in electric motor manufacturing



- A subfield of AI comprises the so-called Semantic Technologies, which play a crucial role in knowledge management.
- Semantic technologies can be concretely utilized in knowledge management through Semantic Wikis. Unlike conventional wikis, Semantic Wikis have a model of the knowledge stored within them. This enables the capture of knowledge about pages and their relations, allowing computers to automatically process at least parts of the knowledge.
- In the currently emerging Semantic Wiki for the design of production systems for electric motors, all process alternatives, their descriptions, and attributes are recorded. As a result, the production planner can automatically generate overviews that compare different process alternatives and highlight known interdependencies with product properties.



Mögliche Verfahren

	Hoher Nutzfaktor	Niedrige Wickelkopfhöhe	Symmetrie der Wicklung	Maximaler Drahtdurchmesser
Einträufeln	100 %	100 %		5 mm
Einziehtechnik	25 %	50 %		2 mm
Flyerwickeln	50 %	75 %	75 %	2.5 mm

Nadelwickeln

Das Nadelwickelverfahren ist ein vielseitiges Wickelverfahren, dass für viele verschiedene Wickelprodukte eingesetzt wird. Es wird für folgende Bauteile eingesetzt:

- Bauteile mit geringen Windungszahlen bei dicker Drahtstärke,
- Innengenuetete Bauteile,
- Bauteile mit hoher Masse,
- Bauteile mit schlechter Zugänglichkeit des Wickelraums,
- Polkettenwicklungen mit mehreren Phasen,
- Ankerwickeln und
- Bauteile mit mehreren Verschaltungsdrähten.^{[1] S. 227}

Nadelwickeln konzentrierter Stator Wicklungen

Mit dieser Verfahrensvariante können innen- und außengetutete Statorpakete mit und ohne Segmentierung bewickelt werden^{[4] S. 59, S. 62}. Der Prozessablauf beginnt mit der Zuführung des Spulenkörpers und der Positionierung der Nadelwickeldüse in Grundstellung. Im zweiten Prozessschritt wird der Spulenkörper in das Aufnahmewerkzeug der Nadelwickelmaschine gespannt. Anschließend wird die Nadeldüse um 90 Grad in ihre senkrechte Stellung geschwenkt, um den Anfangsterminierung des Wickeldrahtes am Kontaktpunkt vorzunehmen. Nach erfolgter Rückfahrt und Anwenkung der Nadeldüse um 90 Grad in die radiale Ausrichtung zum Nutgrund, erfolgt die Positionierung der Nadeldüse über dem ersten innenliegenden Nutschlitz. Anschließend kommt es zur Ausführung des Abwärtschubes im Statorinneren. Die Nadeldüse durchfährt den Nutschlitz in einer Abwärtsbewegung. Der Endpunkt der Aufwärtsbewegung ist an der

Informationen zu Nadelwickeln	
Allgemeine Attribute	
Einmalkosten	50 % [1] S. 174, 264f
Betriebskosten	
Min. Prozesszeit	57,5 s [1] S. 264f
Max. Prozesszeit	120 s [1] S. 264f
Stückzahlflexibilität	
Variantenflexibilität	75 % [1] S. 228
Prozesssicherheit	
Leistungsmerkmale	

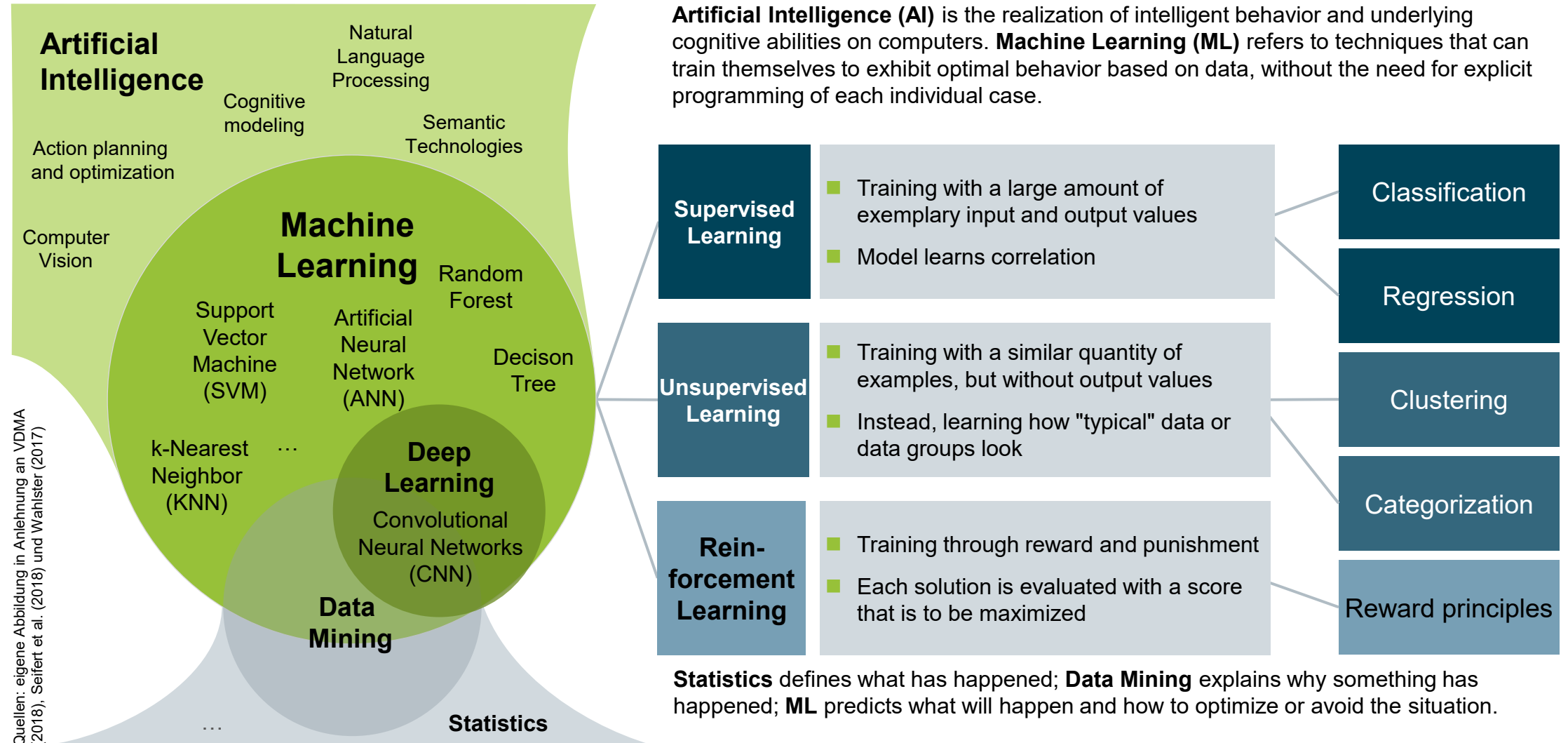
Based on current challenges, approaches to reducing costs, increasing scalability, using Industry 4.0 technologies and recycling are outlined.

- Current challenges and trends in electric motor production
- Cost structures and measures to optimize production costs
- Solution approaches for scalable multi-variant production
- **Potentials of Industry 4.0 and artificial intelligence**
 - Semantic technologies to support the design of production systems
 - **Machine learning for process and production optimization**



Graphic: VW

By using machine learning methods, complex processes can be modeled and valuable knowledge generated without the need for an in-depth understanding of the process.



Quellen: eigene Abbildung in Anlehnung an VDMA (2018), Seifert et al. (2018) und Wahlester (2017)

Based on the fundamental potential of ML, meaningful use cases must be systematically identified, evaluated and implemented in your own company.

Basic use cases of machine learning

Maintenance:

Detection and prediction of the wear condition of plant components reduces plant downtime and tooling costs



Quality management:

Detection or prediction of quality based on process data saves onerous inspection steps or sampling checks



Process monitoring/control:

Processes signal anomalies or even automatically adapt to deviations from upstream steps



Robotics:

AI-based object and pose determination as well as assembly strategies enable the handling of various objects



Production planning and control:

Agent-based, decentralized control enables cycle time- or peak-load-optimized machine occupancy



Identification of specific ML use cases

Opportunities Push | Problem Pull

Specification

Benefit analysis

Effort analysis

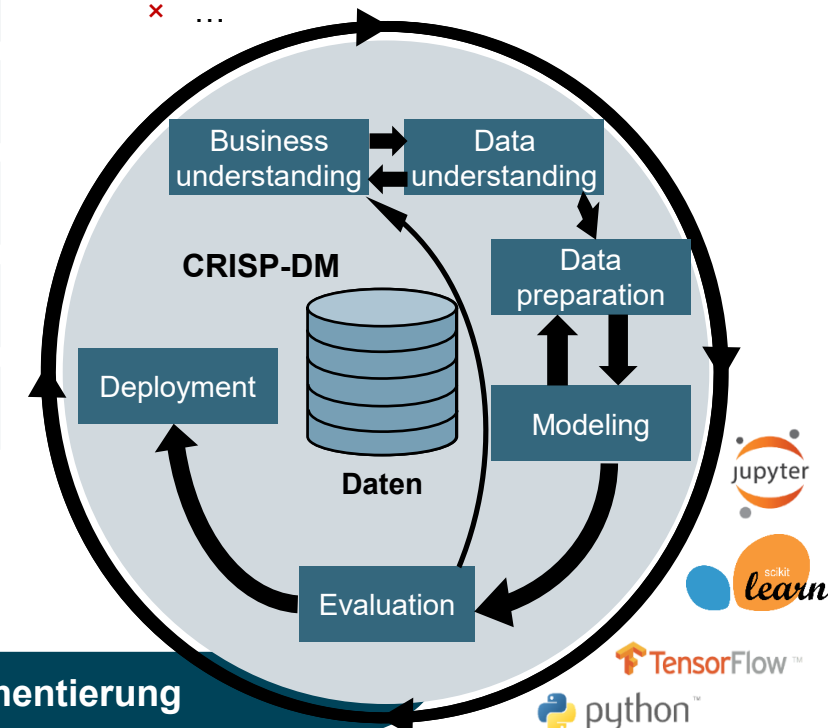
Prioritization and selection

Development of a concrete implementation strategy

Implementierung

Challenges

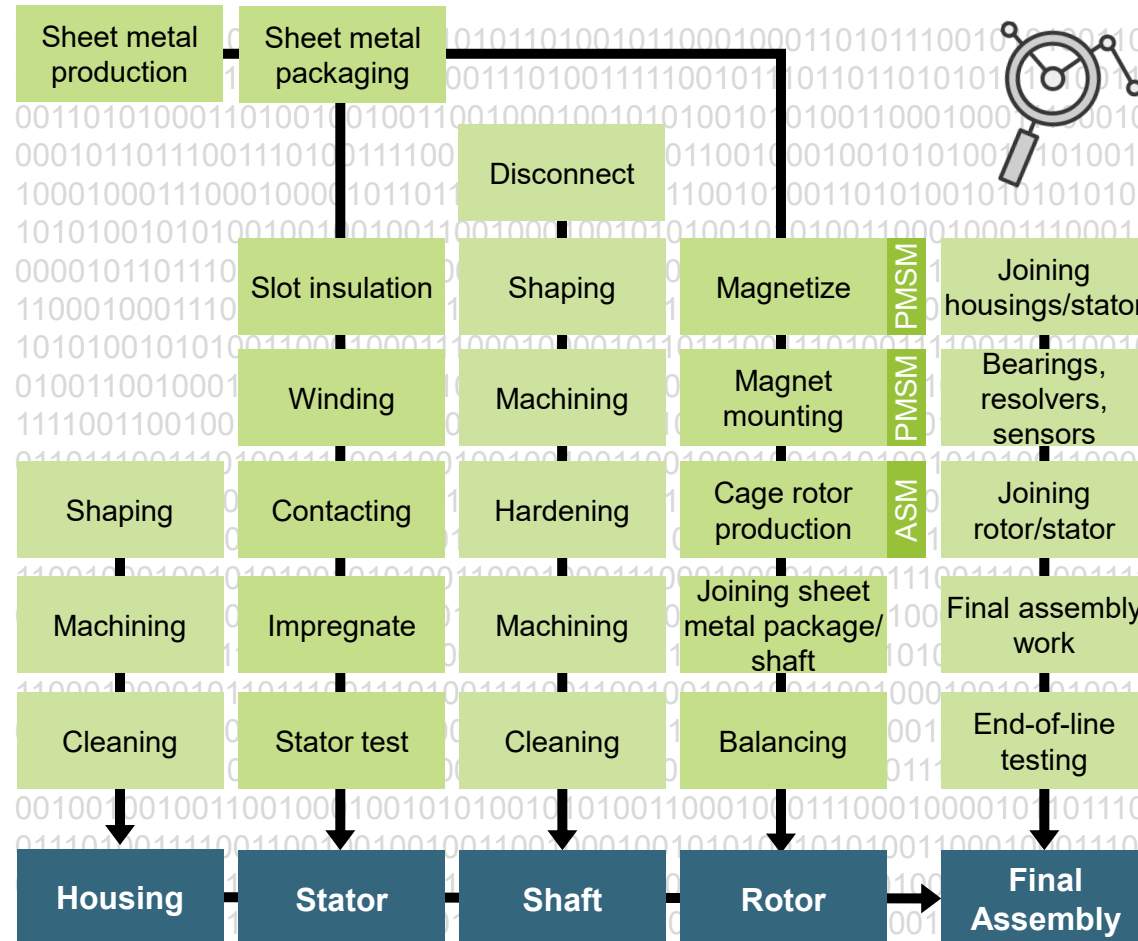
- ✗ Data availability and quality
- ✗ Integration of legacy systems
- ✗ Time-consuming data preparation
- ✗ Quality and interpretability of ML models
- ✗ ...



Source: Mayr, Kißkalt et al. (2019) und Mayr, Seefried et al. (2019)

The application scenarios for ML in electric motor production and the processes involved are diverse, as the list of possible ML use cases shows.

Overview of first, already published use cases of ML in electric motor production



Electric motor production		Selected ML approaches
Individual process steps	Housing and shaft production	Kiðkalt, Mayr et al. (2018) Lutz, Kiðkalt, Mayr et al. (2020) Kiðkalt, Mayr et al. (2020)
	Sheet metal package production	-
	Winding	Mayr, Kiðkalt et al. (2021) Mayr, Röhl et al. (2021) Mayr, Scheffler et al. (2022)
	Contacting	Fleischmann et al. (2016) Mayr, Meyer et al. (2017) Weigelt, Mayr et al. (2018) Mayr, Lutz et al. (2018) Vater et al. (2019) Mayr, Hauck et al. (2020) Raffin et al. (2021)
	Magnet mounting	Mayr, Meyer et al. (2017)
	Final assembly	Meiners, Mayr & Franke (2021)
	Cross-process	Blank et al. (2019) Meiners, Mayr et al. (2019) Meiners, Mayr & Franke (2021)

Legend: Direct or indirect reference to e-motor production

Source: Mayr, Seefried et al. (2019),
Mayr, Weigelt, Masuch et al. (2018) und Mayr, Weigelt, Lindenfels et al. (2018)

A current research project is investigating the extent to which the quality of hairpin welded joints can be predicted on the basis of process data using ML.

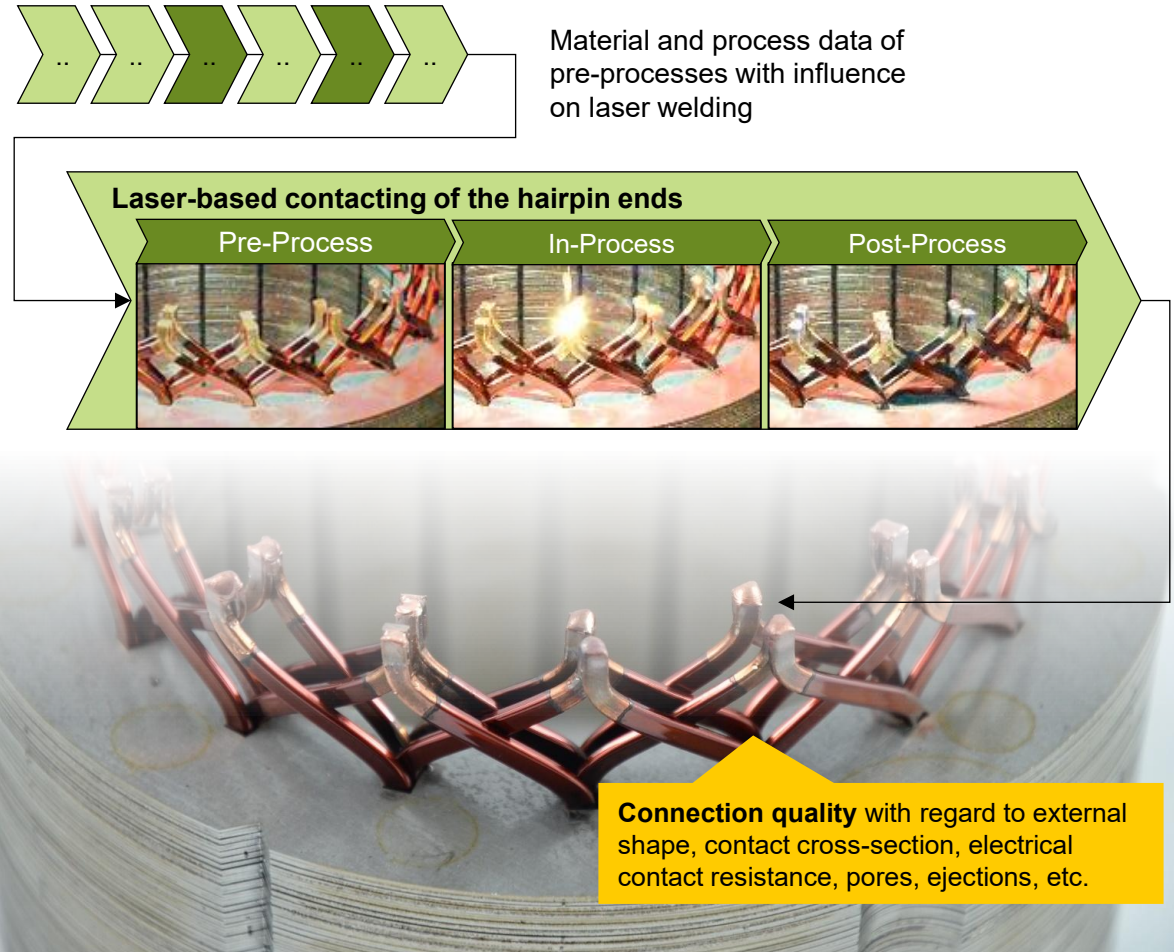
Application example | ML-based quality monitoring and optimization for laser welding of hairpins

Motivation

- Due to the high number of contact points, laser welding plays a key role in hairpin stator production
- In order to realize a stable laser welding process, all influencing variables within the laser welding process itself, but also the sometimes significant interactions with upstream process steps (e.g. cutting to length and stripping) must be taken into account
- In some cases, there is no fast, cost-effective test procedure that can be used to monitor the electrical and mechanical properties of a welded joint inline

Objective

- The extent to which ML can be used to predict the quality of each welded joint based on existing material and process data as well as additional integrable sensor technology (e.g. optical coherence tomography (OCT), see Baader et al. 2021) is therefore currently being investigated
- In the future, the accelerated ML solution should make it possible to detect deviations in upstream process steps at an early stage and compensate for them by intelligently adapting the laser welding process



Initial investigations have already shown that pre- and post-process images can be used to make statements about the joint cross-section.

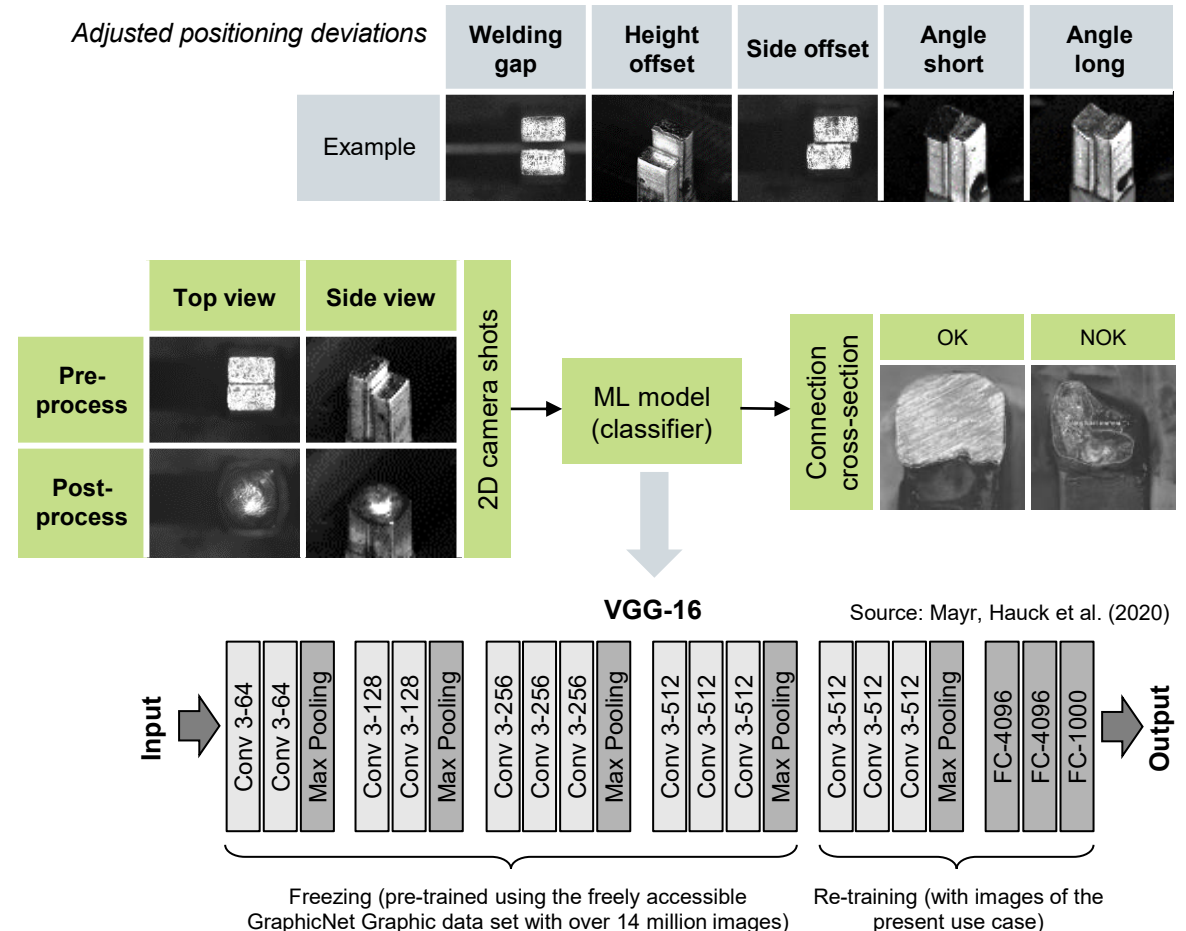
Application example | ML-based quality monitoring and optimization for laser welding of hairpins

Motivation

- Data set of 288 welds with adjusted positioning deviations (welding gap, height and lateral offset)
- Recording of pre-process and post-process 2D Graphics from five different perspectives
- For training data set only: Measurement of the connection cross-section using micrographs
- Question of whether the cross-section of the bond can be estimated using an ML model based on the 2D images and which of the perspectives are particularly relevant for this purpose
- Use of a pre-trained CNN based on the known VGG16 architecture and adaptation of this to the use case by training the last levels (transfer learning)

Results

Based on the pre- and post-process images from the top and side view, the trained CNN can identify 100% of the welds that have an insufficient connection cross-section; however, due to the small training data set, the high recall is accompanied by some pseudo errors



In hairpin bending, it has already been shown that geometric features such as the resulting leg spacing can be predicted using process curves.

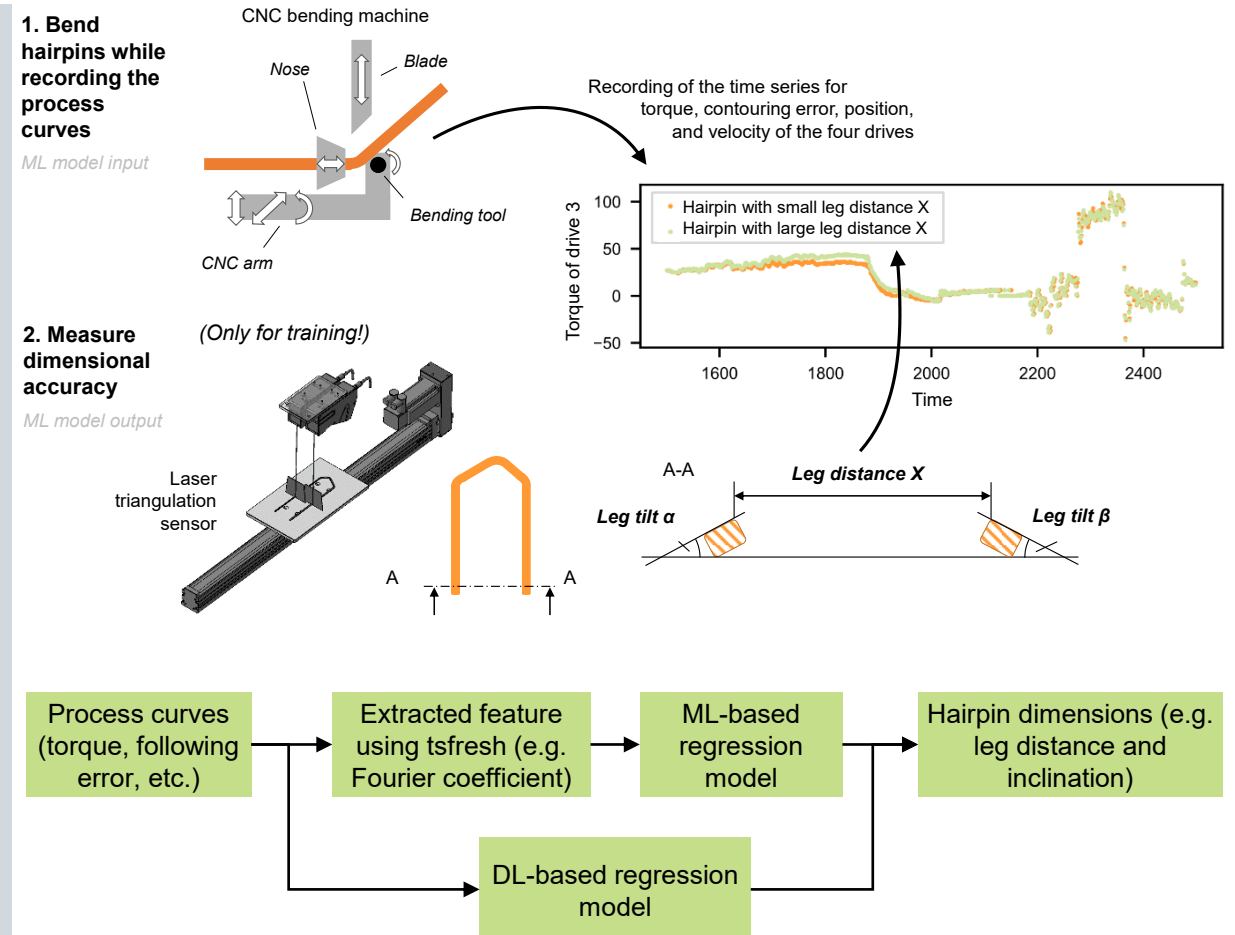
Application example | Data-driven quality monitoring based on process curves for hairpin bending

Motivation and objective

- Although forming processes such as bending are strongly influenced by material tolerances, the quality of bent hairpins is usually not checked inline, but only randomly
- The aim is therefore to investigate whether the dimensional accuracy of a bent hairpin can be estimated on the basis of process data available from the bending machine

Approach and results

- For an initial proof of concept, a bending data set was created in the laboratory environment of an industrial partner (672 samples)
- In order to generate the labels required for training, each hairpin was geometrically measured after bending
- When training different multi-output regressors, not only feature-based ML approaches but also a DL approach were investigated, with the latter performing best
- 1D Convolutional Neural Networks (CNNs) eliminate the need for manual feature engineering and extract relevant features independently. Using a 1D-CNN, leg distance can be determined from torque and tracking error curves with an R^2 score of 0.93.



Source: Mayr, Röhl et al. (2021)

The contents of the lecture unit can be further deepened with the following specialized literature:

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- Lomakin, A., Mayr, A., Graichen, K. & Franke, J. (2020). Optimization of Direct Winding Processes Based on a Holistic Control Approach. In 2020 10th International Electric Drives Production Conference (EDPC) (pp. 1–7). IEEE. <https://doi.org/10.1109/EDPC51184.2020.9388184>
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FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

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